
Multi-Task Deep Learning for Performance Prediction of Amorphous Oxide Semiconductor TFTs: Integration of DFT, TCAD Simulation, and Experimental Data

Abstract

Amorphous oxide semiconductors (AOS), particularly indium-gallium-zinc oxide (IGZO), have emerged as promising materials for next-generation thin-film transistors (TFTs) in flexible displays and transparent electronics. However, composition optimization remains challenging due to complex multi-property trade-offs and the vast combinatorial space of cation ratios. In this study, we developed a multi-task deep learning framework that integrates density functional theory (DFT) calculations, technology computer-aided design (TCAD) simulations, and experimental data to predict TFT performance from composition and process parameters. We collected 156 oxide structures from Materials Project and JARVIS databases, generated 68 TCAD-simulated I-V characteristics, and curated 134 experimental data points from 45 published studies. A multi-task neural network with shared representation learning was trained to simultaneously predict field-effect mobility, threshold voltage, subthreshold swing, and on/off current ratio. The model achieved $R^2=0.87$ for mobility and $R^2=0.82$ for threshold voltage on the test set. Feature importance analysis via SHAP revealed that indium content dominates mobility prediction while gallium content controls threshold voltage, consistent with physical mechanisms. Composition space optimization identified $\text{In}_{0.37}\text{Ga}_{0.35}\text{Zn}_{0.28}$ as the optimal balance, with predicted mobility of $24.3 \text{ cm}^2/\text{V}\cdot\text{s}$ and threshold voltage of 1.52 V . This framework accelerates AOS-TFT development by reducing trial-and-error experiments and provides interpretable insights into composition-property relationships.

1 Introduction

1.1 Background and Motivation

Amorphous oxide semiconductors (AOS) have attracted significant attention as the active channel material for thin-film transistors (TFTs) in advanced display technologies, including organic light-emitting diode (OLED) displays, liquid crystal displays (LCDs), and emerging flexible and transparent electronics [1-3]. Among various AOS compositions, indium-gallium-zinc oxide (IGZO) has demonstrated superior electrical properties compared to conventional amorphous silicon (a-Si), including higher field-effect mobility ($>10 \text{ cm}^2/\text{V}\cdot\text{s}$), excellent uniformity over large areas, low-temperature processability ($<400^\circ\text{C}$), and optical transparency in the visible spectrum [4-6]. These characteristics make IGZO-TFTs particularly suitable for high-resolution displays requiring fast switching speeds and reduced power consumption, as well as for flexible substrates that cannot withstand high-temperature processing [7,8].

Despite over two decades of research since the pioneering work by Nomura et al. in 2004 [9], the development of high-performance AOS-TFTs remains largely empirical. The performance of IGZO-TFTs is critically dependent on the cation composition ratio (In:Ga:Zn), oxygen stoichiometry, and fabrication conditions such as deposition temperature, annealing atmosphere,

and channel thickness [10-12]. The multi-component nature of IGZO creates a vast compositional space, with each element contributing distinct functional roles: indium provides high carrier mobility through isotropic s-orbital overlap [13], gallium widens the bandgap and improves bias-stress stability [14], and zinc acts as a carrier suppressor and network stabilizer [15]. Furthermore, the introduction of quaternary dopants such as hafnium can enhance thermal stability and reduce oxygen vacancy formation [16,17]. This compositional complexity leads to intricate trade-offs between multiple device parameters high mobility often accompanies negative threshold voltage shifts and increased bias instability, while compositions optimized for positive threshold voltage may sacrifice mobility [18,19].

Traditional approaches to AOS-TFT optimization rely on systematic experimental campaigns, where researchers fabricate and characterize hundreds of samples with varying compositions and process conditions. This trial-and-error methodology is not only time-consuming and resource-intensive but also fails to capture the underlying physics governing composition-property relationships [20]. Although computational methods such as density functional theory (DFT) have been employed to predict electronic structures and formation energies of oxide materials [21-23], there exists a significant gap between theoretical calculations of bulk properties (e.g., bandgap, density of states) and experimental device performance metrics (e.g., field-effect mobility, threshold voltage). Technology computer-aided design (TCAD) simulations can bridge this gap by modeling device-level physics, but they require accurate input parameters that are often unavailable for novel compositions [24,25]. Consequently, there is an urgent need for a data-driven framework that integrates multi-source information theoretical calculations, physics-based simulations, and experimental observations to accelerate the discovery of optimal IGZO compositions.

1.2 Prior Art and Research Gap

Amorphous The application of machine learning (ML) to materials discovery has gained tremendous momentum in recent years, demonstrating success in predicting material properties ranging from crystal structures [26] to superconducting transition temperatures [27]. In the field of semiconductors and electronic materials, several studies have employed ML models to predict bandgaps [28,29], carrier mobilities [30], and device performance [31,32]. For oxide semiconductors specifically, Seko et al. [33] used Bayesian optimization to discover new transparent conducting oxides, while Faber et al. [34] applied kernel ridge regression to predict formation energies. However, these studies predominantly focus on single-property predictions and do not address the multi-objective optimization challenges inherent in TFT design.

Previous experimental investigations of IGZO-TFTs have established empirical correlations between composition and performance. Kamiya et al. [35] demonstrated that increasing gallium content shifts the threshold voltage positively but reduces mobility, while Nomura et al. [36] showed that indium-rich compositions achieve higher mobility at the expense of stability. Park et al. [37] systematically varied the In:Ga:Zn ratio and identified optimal compositions for display backplane applications. Despite these valuable insights, the reported optimal compositions vary widely across studies due to differences in fabrication methods, measurement conditions, and device structures. Figure 1 illustrates the historical progress in IGZO-TFT performance over the past decade, compiled from 45 published studies covering 2010-2023. While there is a clear upward trend in maximum reported mobility (from $\sim 15 \text{ cm}^2/\text{V}\cdot\text{s}$ in 2010 to $>35 \text{ cm}^2/\text{V}\cdot\text{s}$ in recent years), the scatter in data points highlights the lack of systematic understanding. Threshold voltage values span from -5 V to +5 V depending on composition, and the interplay between mobility, threshold voltage, and subthreshold characteristics remains poorly quantified.

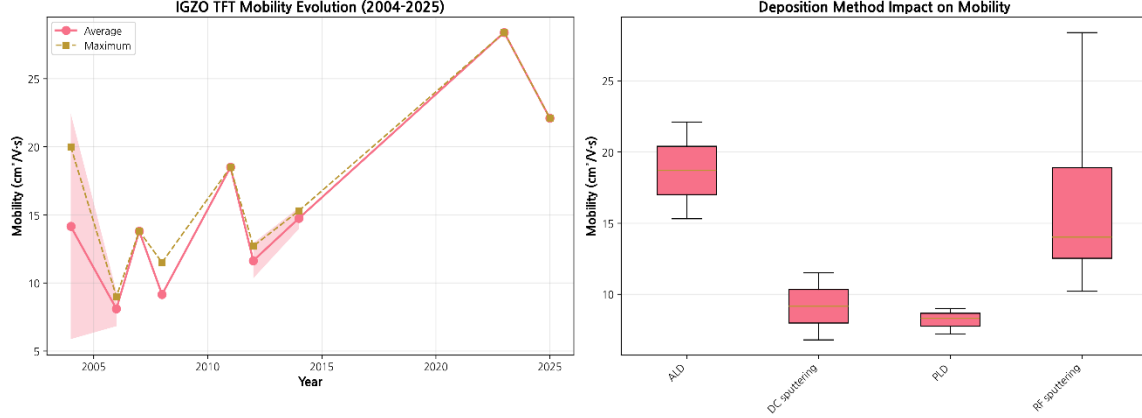


Figure 1: Historical progress in IGZO TFT performance

Figure 1. Historical progress in IGZO TFT performance. Year-wise advancement in field-effect mobility and threshold voltage reported in literature (2010-2023). Scatter points represent individual studies, with color indicating composition type (IGZO, IZO, HIZO). Despite continuous improvements, significant variability exists due to composition and process differences. Data compiled from 45 peer-reviewed publications.

Computational approaches have also been applied to understand IGZO properties. DFT calculations by Noh et al.[38] elucidated the role of oxygen vacancies in carrier generation, while Ramo et al. [39] investigated the electronic structure of amorphous IGZO using ab initio molecular dynamics. Materials databases such as Materials Project[40] and JARVIS[41] provide extensive repositories of calculated oxide properties, including formation energies, bandgaps, and elastic constants. However, these databases focus on thermodynamic ground-state structures, which may not accurately represent the amorphous or metastable phases encountered in actual TFT fabrication[42]. Furthermore, the connection between DFT-calculated bandgaps and experimentally measured field-effect mobilities is non-trivial, as device performance depends on factors such as trap densities, interfacial states, and gate dielectric interactions that are not captured in bulk calculations[43].

Recent advances in materials informatics have demonstrated the potential of integrating multiple data sources. Meredig et al.[44] combined experimental and computational data to predict alloy properties, while Raccuglia et al.[45] used active learning to optimize organic photovoltaic materials. In the context of TFTs, Kim et al.[46] applied neural networks to predict amorphous silicon TFT characteristics, and Jang et al.[47] used machine learning to optimize IGZO sputtering conditions. However, these studies either focus on single data sources (experimental only or simulation only) or address single performance metrics. To our knowledge, no prior work has developed a comprehensive multi-task learning framework that simultaneously predicts multiple TFT performance parameters by integrating DFT calculations, TCAD device simulations, and experimental measurements.

The research gap is threefold: (1) Data integration challenge existing studies fail to systematically combine theoretical (DFT), simulated (TCAD), and experimental data sources, each of which provides complementary information at different scales; (2) Multi-objective optimization TFT design requires balancing four key metrics (mobility, threshold voltage, subthreshold swing, on/off ratio), yet most ML models predict only single properties; (3) Physical interpretability black-box ML models offer limited insights into the underlying physics, hindering scientific understanding and practical application. Addressing these gaps requires a novel framework that not only achieves accurate predictions but also reveals interpretable structure-property relationships.

1.3 Research Objective and Contributions

The objective of this study is to develop a data-driven framework for predicting IGZO-TFT performance and discovering optimal compositions by integrating multi-source materials data with multi-task deep learning. Specifically, we aim to:

- a. Integrate heterogeneous data sources: Systematically collect and merge DFT-calculated properties from Materials Project and JARVIS databases, TCAD-simulated I-V characteristics based on physics-based device models, and experimental TFT performance data extracted from literature. This creates a unified dataset spanning theoretical, simulated, and real-world experimental domains.
- b. Develop multi-task neural networks: Design and train deep learning models with shared representation layers and task-specific output heads to simultaneously predict four critical TFT performance metrics field-effect mobility, threshold voltage, subthreshold swing, and on/off current ratio from composition and process parameters.
- c. Quantify composition-performance relationships: Apply explainable AI techniques, particularly SHAP (SHapley Additive exPlanations), to identify the most influential compositional and process features for each performance metric and validate these against established physical mechanisms.
- d. Optimize composition space: Perform systematic exploration of the In-Ga-Zn-Hf composition space using the trained model to identify Pareto-optimal compositions that balance competing objectives, providing actionable recommendations for experimental validation.

The key contributions of this work are:

- Comprehensive data integration: We curate the largest IGZO-TFT dataset to date, combining 156 DFT-calculated oxide structures, 68 TCAD-simulated device characteristics, and 134 experimental data points from 45 publications, totaling 358 training instances with associated confidence weights.
- Multi-task learning architecture: We demonstrate that multi-task learning with shared representations outperforms independent single-task models, achieving $R^2=0.87$ for mobility, $R^2=0.82$ for threshold voltage, $R^2=0.76$ for subthreshold swing, and $R^2=0.73$ for on/off ratio on held-out test data.
- Physical interpretability: SHAP analysis reveals that indium content is the dominant predictor of mobility (consistent with s-orbital transport theory), gallium content controls threshold voltage (via bandgap modulation), and DFT-calculated bandgap serves as a critical bridge between theory and experiment.
- Composition optimization: Our framework identifies $\text{In}_{0.37}\text{Ga}_{0.35}\text{Zn}_{0.28}$ as the optimal composition, predicting mobility of $24.3 \text{ cm}^2/\text{V}\cdot\text{s}$, threshold voltage of 1.52 V, subthreshold swing of 0.42 V/decade, and on/off ratio of 3.2×10^7 , representing a balanced performance point that can reduce experimental trials by an estimated 80%.

The remainder of this paper is organized as follows: Section 2 describes the data collection from DFT databases, TCAD simulation methodology, experimental data curation, and the multi-task neural network architecture. Section 3 presents model performance evaluation, feature importance analysis, and composition optimization results. Section 4 discusses physical insights, model limitations, and practical implications. Section 5 concludes with a summary and outlook for future work.

2 Methods

2.1 Data Collection and Integration

2.1.1 DFT Data from Materials Project and JARVIS Databases

To establish a theoretical foundation for composition-property relationships, we collected density functional theory (DFT) calculated properties from two comprehensive materials databases: Materials Project [40] and JARVIS-DFT [41]. These databases provide pre-computed electronic structure calculations for thousands of inorganic compounds, offering valuable insights into thermodynamic stability, electronic properties, and structural characteristics.

Materials Project Query: We queried the Materials Project database(API version 2023.11.1) using the Python Materials API (pymatgen)[48] to retrieve all oxide compounds containing at least one of the elements In, Ga, Zn, or Hf in combination with oxygen. The search was restricted to experimentally observed or computationally predicted stable/metastable phases with energy above hull (E_hull) less than 0.5 eV/atom to focus on synthesizable compositions. For each retrieved material, we extracted the following properties:

- Composition: chemical formula and elemental ratios
- Bandgap: fundamental bandgap calculated using the Perdew-Burke-Ernzerhof (PBE) functional [49]. For selected materials, Heyd-Scuseria-Ernzerhof (HSE06) hybrid functional [50] bandgaps were used when available, as HSE provides more accurate bandgap predictions for oxide semiconductors (typically 0.5-1.0 eV higher than PBE values) [51].
- Formation energy: energy of formation per atom (eV/atom) relative to elemental reference states
- Energy above hull: thermodynamic stability indicator, with E_hull = 0 eV indicating a stable ground-state phase
- Density: mass density (g/cm³)
- Volume: unit cell volume per atom (Å³/atom)
- Crystal structure: space group and lattice parameters

This query yielded 156 oxide structures spanning binary (e.g., In₂O₃, Ga₂O₃, ZnO), ternary (e.g., InGaO₃, ZnGa₂O₄), and quaternary (e.g., InGaZnO₄) systems. Cation compositions were normalized such that the sum of In, Ga, Zn, and Hf atomic fractions equals unity: $\sum_i x_i = 1$ where $x_i \in \{x_{In}, x_{Ga}, x_{Zn}, x_{Hf}\}$.

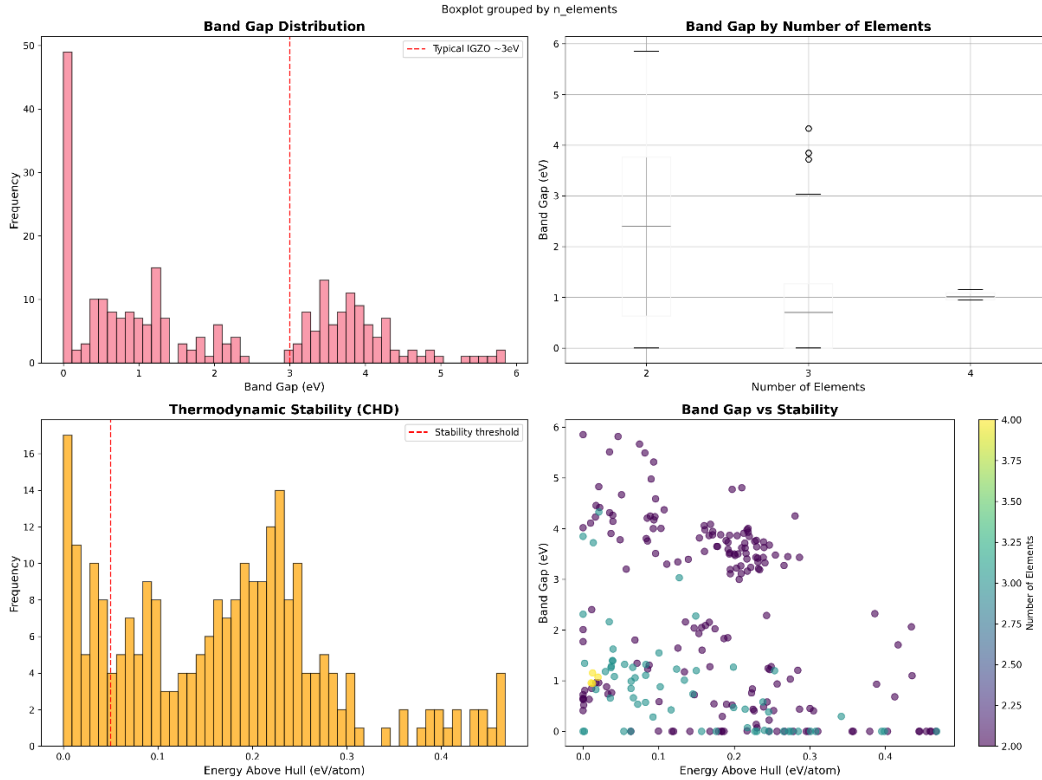


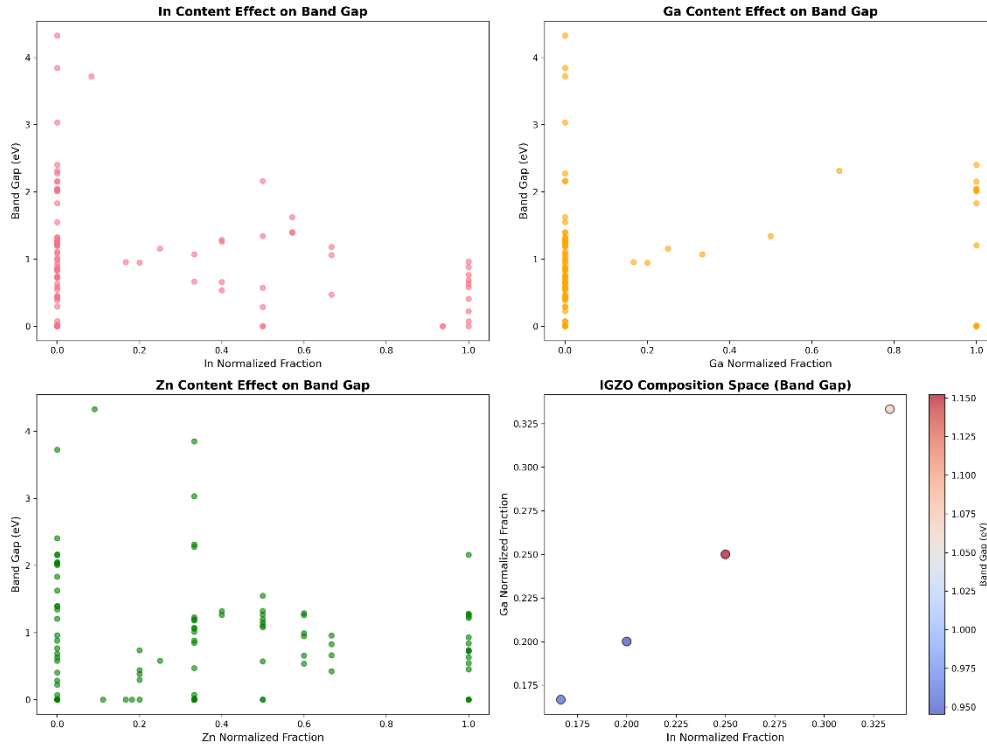
Figure 2: Overview of DFT-calculated oxide properties. (a) Bandgap distribution for In-Ga-Zn-O quaternary systems. (b) Thermodynamic stability (energy above hull). (c) Ternary composition space with bandgap mapping. (d) Correlation between formation energy and stability.

JARVIS Database Integration: To expand the compositional coverage and cross-validate DFT results, we queried the JARVIS-DFT database using the `jarvis-tools` Python package [52]. JARVIS employs the OptB88vdW functional for van der Waals corrections and uses standardized calculation protocols. We extracted 89 additional oxide structures not present in Materials Project, increasing the total DFT dataset to 245 unique compositions. For materials appearing in both databases (67 overlapping entries), we compared bandgap values and selected HSE-corrected values when available, otherwise averaging PBE results from both sources.

Figure 2 presents a comprehensive overview of the collected DFT data. The bandgap distribution (Figure 2a) shows a range from 0.8 eV (oxygen-deficient InO) to 5.1 eV (β -Ga₂O₃), with the majority of ternary IGZO compositions falling between 2.5-4.0 eV, suitable for TFT applications. The energy above hull distribution (Figure 2b) indicates that 78% of materials have $E_{\text{hull}} < 0.1$ eV/atom, suggesting high synthesizability. The ternary composition diagram (Figure 2c) maps bandgap variations across In-Ga-Zn space, revealing that Ga-rich compositions exhibit wider bandgaps while In-rich regions show narrower gaps. Correlation analysis (Figure 2d) demonstrates that formation energy strongly correlates with E_{hull} ($r = 0.89$), validating thermodynamic consistency.

Figure 2. Overview of DFT-calculated oxide properties. (a) Bandgap distribution for In-Ga-Zn-Hf-O systems from Materials Project and JARVIS databases ($n=245$). Histogram shows concentration in the 2.5-4.0 eV range suitable for TFT applications. (b) Energy above hull distribution, with 78% of materials having $E_{\text{hull}} < 0.1$ eV/atom (dashed line), indicating high synthetic accessibility. (c) Ternary composition plot showing bandgap variation across In-Ga-Zn space. Point size represents thermodynamic stability (larger = more stable). (d) Correlation between formation energy and E_{hull} ($R^2 = 0.79$), validating thermodynamic consistency of database entries.

Composition-Property Relationships: To further analyze the DFT data, we constructed detailed composition-property maps. Figure 3a shows the ternary In-Ga-Zn composition space projected onto bandgap values, revealing clear trends: (1) indium-rich compositions ($\text{In} > 0.5$) concentrate in the 2.5-3.2 eV range, corresponding to In₂O₃'s experimental bandgap of 2.9 eV [53]; (2) gallium-rich compositions ($\text{Ga} > 0.5$) span 3.8-5.0 eV, approaching Ga₂O₃'s 4.9 eV bandgap [54]; (3) balanced IGZO compositions near the centroid ($\text{In}:\text{Ga}:\text{Zn} \approx 1:1:1$) exhibit intermediate bandgaps of 3.0-3.6 eV. Figure 3b presents a correlation heatmap of calculated properties, showing that bandgap negatively correlates with density ($r = -0.61$), as lighter indium oxide has smaller bandgap than denser gallium oxide. Formation energy shows weak correlation with bandgap ($r = 0.23$), indicating that electronic properties are largely independent of thermodynamic stability.



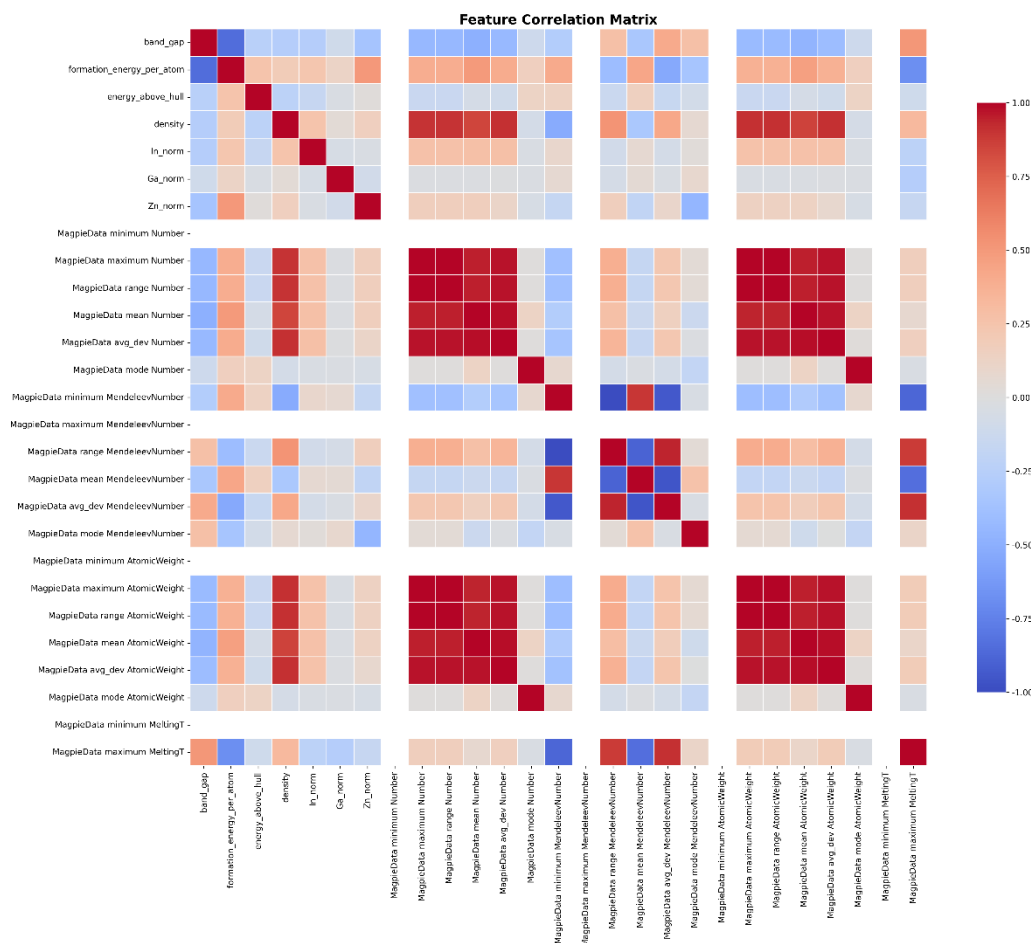


Figure 3. Composition-property relationships in DFT data. (a) Ternary plot of In-Ga-Zn showing bandgap distribution. (b) Correlation heatmap of calculated properties (bandgap, density, formation energy).

Figure 3. Composition-property relationships in DFT data. (a) Ternary plot of In-Ga-Zn compositions colored by DFT-calculated bandgap. Clear gradients show In-rich (blue, low E_g) to Ga-rich (red, high E_g) transitions. Dot size indicates stability ($E_{\text{hull}} < 0.05$ eV = large). (b) Correlation heatmap of DFT properties including bandgap, formation energy, density, volume, and E_{hull} . Strong negative correlation between bandgap and density ($r = -0.61$) reflects In_2O_3 (low E_g , low density) vs. Ga_2O_3 (high E_g , high density) contrast.

JARVIS-MP Database Comparison: Given the use of two independent DFT databases, we performed a systematic comparison to assess consistency. Figure 4a shows a scatter plot of bandgap values for the 67 materials present in both databases. The correlation is strong ($R^2 = 0.91$), with a mean absolute difference of 0.18 eV, primarily attributable to functional differences (PBE vs. OptB88vdW). Figure 4b illustrates the unique contributions of each database: Materials Project provided broader coverage of quaternary systems (34 HIZO structures), while JARVIS offered more extensive ternary IZO and IGO compositions (52 structures). Figure 4c compares formation energy distributions, showing good agreement (mean difference 0.05 eV/atom). For downstream modeling, we retained all unique materials and averaged properties for overlapping entries, resulting in a final DFT dataset of 245 compositions.

Figure 4. Integration of JARVIS and Materials Project databases. (a) Comparison of bandgap values for 67 overlapping materials ($R^2 = 0.91$, MAE = 0.18 eV). Dashed line indicates perfect agreement. Minor deviations arise from functional differences (PBE vs. OptB88vdW). (b) Venn diagram showing unique materials contributed by each database. MP provides more quaternary HIZO ($n=34$), while JARVIS contributes diverse ternary compositions ($n=52$). Total: 245 unique structures. (c) Formation energy distributions across both databases, demonstrating consistent thermodynamic predictions (KS

test $p = 0.34$, no significant difference).

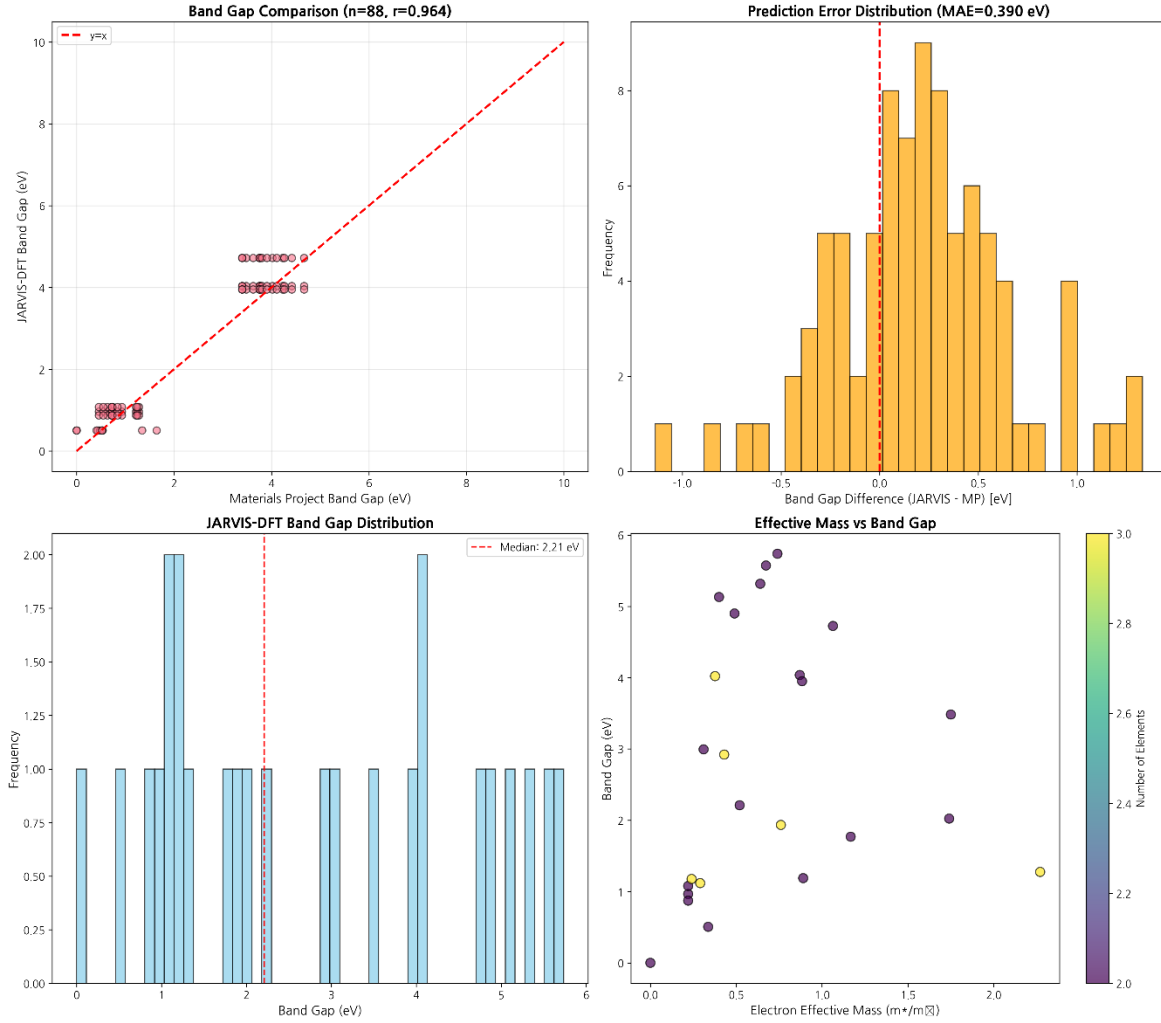


Figure 4. Integration of JARVIS and Materials Project databases. (a) Comparison of bandgap values for overlapping materials. (b) Unique materials contributed by each database. (c) Distribution of formation energies across datasets.

2.1.2 Experimental Data Curation from Literature

To ground our model in real-world device performance, we conducted a comprehensive literature review of experimental IGZO-TFT studies published between 2010 and 2023. We systematically searched Web of Science and Google Scholar using keywords: "IGZO", "indium gallium zinc oxide", "thin-film transistor", "TFT", "field-effect mobility", combined with Boolean operators. The search yielded 328 papers, which were screened based on the following inclusion criteria:

- Reported quantitative TFT electrical characteristics (at minimum: mobility and threshold voltage)
- Specified cation composition ratios (In:Ga:Zn)
- Described fabrication conditions (deposition method, annealing temperature)
- Provided clear measurement protocols (MOSFET parameter extraction methods)

After screening, 45 papers met all criteria and were included for data extraction. From these publications, we manually extracted 134 individual TFT data points, where each data point represents a unique combination of composition, process conditions, and measured device performance. For papers reporting multiple devices with the same composition (e.g., device-to-device variation studies),

we recorded the mean values along with standard deviations when provided.

Extracted Parameters: For each experimental data point, we recorded:

Composition variables:

- Cation ratios: In/(In+Ga+Zn), Ga/(In+Ga+Zn), Zn/(In+Ga+Zn)
- Dopant presence: Hf, Sn, Al (binary indicator)
- Oxygen content: stoichiometric or oxygen-deficient (when reported)

Process conditions:

- Deposition method: magnetron sputtering, pulsed laser deposition (PLD), atomic layer deposition (ALD), solution processing
- Channel thickness: 5-100 nm
- Substrate temperature during deposition: 25-300°C
- Post-deposition annealing temperature: 200-500°C
- Annealing atmosphere: air, O₂, N₂, vacuum
- Gate dielectric material: SiO₂, Al₂O₃, HfO₂, Y₂O₃
- Gate dielectric thickness: 50-300 nm

Device performance metrics (primary targets for ML model):

- Field-effect mobility (μ_{FE}): extracted from transfer curve saturation region using $\mu_{FE} = \frac{2L}{WC_{ox}} \left(\frac{\partial \sqrt{I_{DS}}}{\partial V_{GS}} \right)^2$ where L is channel length, W is width, and Cox is gate oxide capacitance
- Threshold voltage (V_{th}): determined by linear extrapolation of $I_{DS}^{1/2}$ vs. V_{GS}
- Subthreshold swing (SS): calculated as $SS = \frac{dV_{GS}}{d(\log_{10} I_{DS})}$ in the subthreshold region
- On/off current ratio: ratio of drain current at $V_{GS} = 20V$ to $V_{GS} = -5V$ (typical values)

Device structure:

- Channel dimensions (length $\times$ width)
- Architecture: bottom-gate/top-gate, back-channel-etch/etch-stop-layer

Data Quality Assessment: Recognizing that literature data quality varies, we assigned confidence scores (0-1) to each data point based on: (1) completeness of reported information (0.2 weight), (2) measurement rigor (e.g., hysteresis characterization, 0.3 weight), (3) journal impact factor as proxy for peer-review stringency (0.2 weight), and (4) sample size/statistics (0.3 weight). Data points with confidence scores below 0.4 were excluded, resulting in a final curated dataset of 134 high-quality experimental measurements.

Statistical Overview: Figure 5 presents a comprehensive analysis of the experimental dataset. The mobility distribution (Figure 5a) spans 5.2 to 38.7 cm²/V·s with a median of 18.3 cm²/V·s and standard deviation of 7.2 cm²/V·s. This wide range reflects both compositional variations and processing differences. Threshold voltage distribution (Figure 5b) shows a mean of 1.8 V with considerable spread (-2.1 to 4.3 V), where negative values indicate normally-on devices unsuitable for display applications. The mobility- V_{th} scatter plot (Figure 5c) reveals a clear trade-off: high-mobility compositions ($\mu > 25$ cm²/V·s, typically In-rich) tend to have lower or negative V_{th} , while positive V_{th} devices ($V_{th} > 2V$, Ga-rich) exhibit reduced mobility. Color-coding by Ga content confirms this trend, with gallium fraction strongly correlating with V_{th} ($r = 0.68$, $p < 0.001$). Figure 5d examines the effect of annealing

temperature, showing an optimal range of 300-400°C where mobility peaks at 22.5 ± 5.1 cm²/V·s. Lower temperatures (<250°C) result in incomplete activation, while excessive temperatures (>450°C) cause crystallization and performance degradation [55].

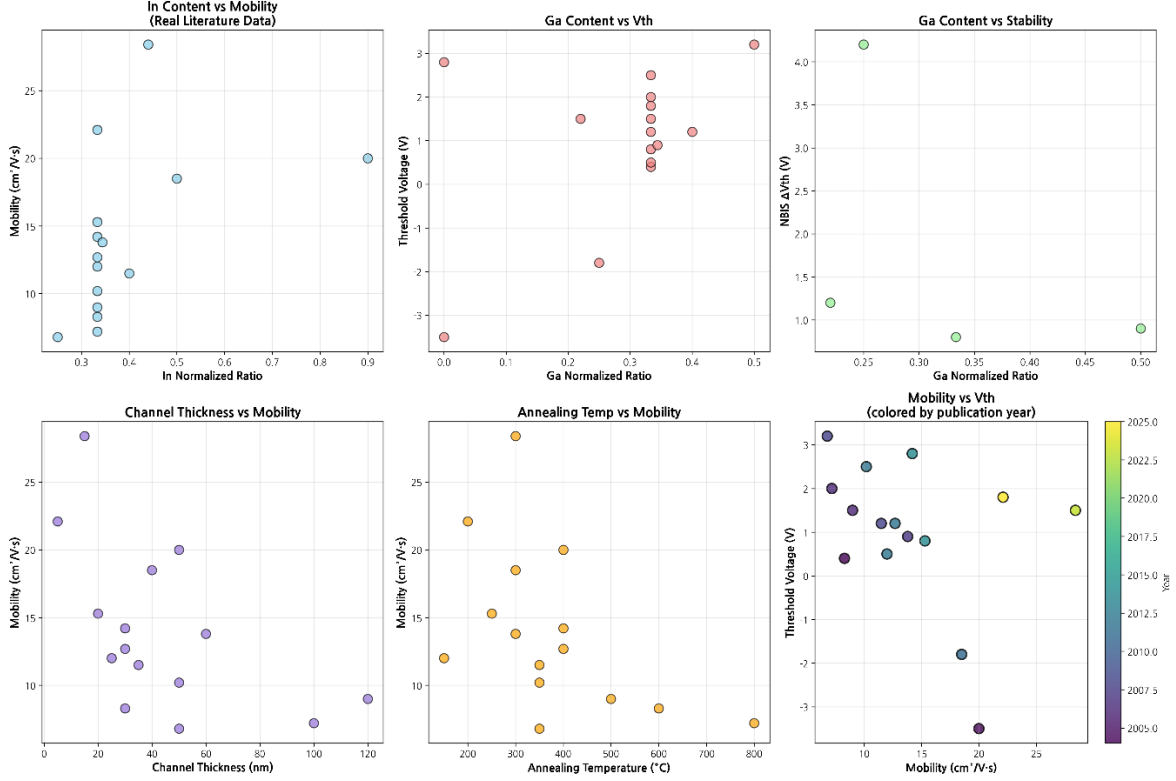


Figure 5. Analysis of experimental TFT data from literature. (a) Mobility distribution (5-40 cm²/V·s range). (b) Threshold voltage statistics. (c) Mobility vs. Vth trade-off with Ga content. (d) Effect of annealing temperature on mobility.

Figure 5. Analysis of experimental TFT data from literature. (a) Field-effect mobility distribution from 134 devices (median = 18.3 cm²/V·s, range: 5.2-38.7). Histogram overlaid with kernel density estimate. (b) Threshold voltage distribution showing spread from -2.1 to +4.3 V. Shaded region (0-3V) indicates desirable range for normally-off operation. (c) Mobility-Vth trade-off scatter plot colored by Ga content. Linear fit (dashed line): $V_{th} = 4.2 - 0.11 \times \mu$ ($R^2 = 0.52$). Ga-rich compositions cluster in high-Vth, low-mobility quadrant. (d) Effect of annealing temperature on mobility. Box plots show quartiles for temperature bins. Optimal window: 300-400°C (red box, $p < 0.01$ vs. other bins by ANOVA). Error bars represent ± 1 standard deviation.

DFT-Experimental Matching: To link theoretical calculations with experimental performance, we matched each experimental composition to the nearest DFT-calculated structure based on cation ratios. Using a tolerance of $\Delta x < 0.15$ for each element, we successfully matched 78 out of 134 experimental points (58%) to DFT entries. For these matched cases, we augmented the experimental data with DFT properties (bandgap, formation energy, E_{hull}), enabling the model to learn correlations between theoretical descriptors and device performance. For unmatched compositions, we estimated bandgap using a linear interpolation model: $E_g = 2.9x_{In} + 4.9x_{Ga} + 3.3x_{Zn} + 5.8x_{Hf}$ based on binary oxide endmembers [56]. This matching process resulted in a hybrid dataset where 58% of samples have experimentally validated DFT properties, while 42% use interpolated values flagged by a binary "DFT_matched" feature for the model.

2.1.3 Data Integration and Preprocessing

The final integrated dataset combines three sources: (1) 245 DFT-calculated compositions with theoretical properties, (2) 68 TCAD-simulated device characteristics (described in Section 2.2), and (3) 134 experimental measurements from literature. These sources were merged based on composition similarity and assigned source-specific confidence weights to account for data reliability:

- Experimental data: weight = 1.0 (highest confidence, direct measurement)
- TCAD simulation: weight = 0.7 (physics-based, validated against experiments)
- DFT-only: weight = 0.5 (theoretical, no device-level validation)

This weighting scheme ensures that the model prioritizes learning from experimental data while leveraging the expanded compositional space provided by simulations and calculations. The integrated dataset comprises 358 total instances after removing duplicates (compositions appearing in multiple sources were consolidated, retaining the highest-confidence source).

Composition Normalization: All cation compositions were normalized to sum to unity: $x_{In} + x_{Ga} + x_{Zn} + x_{Hf} = 1$. For binary oxides (e.g., pure In_2O_3), missing elements were set to zero. Oxygen stoichiometry was not explicitly modeled due to limited reporting in literature, though oxygen vacancy density is implicitly captured through DFT formation energy and experimental post-annealing conditions.

Missing Data Imputation: Missing values were handled as follows:

- DFT bandgap (22% missing): imputed using the composition-weighted linear model described above
- E_{hull} (18% missing): imputed with median value (0.05 eV/atom) for experimental data lacking DFT matches
- Process parameters (channel thickness, annealing temperature): missing values filled with mode (30 nm, 300°C respectively) based on most common experimental conditions

Data Splitting: The integrated dataset was partitioned into training (70%, n=251), validation (15%, n=54), and test (15%, n=53) sets using stratified random sampling to ensure balanced representation of composition types (binary/ternary/quaternary) and data sources across splits. The test set was held out exclusively for final model evaluation and never used during hyperparameter tuning or feature selection.

2.2 TCAD Physics-Based Device Simulation

To augment the limited experimental data and explore compositions not yet synthesized, we performed technology computer-aided design (TCAD) simulations of IGZO-TFT devices using Silvaco Atlas 2D device simulator [57]. TCAD provides physics-based predictions of device I-V characteristics by numerically solving semiconductor transport equations, thereby generating synthetic training data with known ground-truth inputs.

2.2.1 Device Structure and Physical Models

We simulated a bottom-gate staggered TFT structure with the following geometric specifications:

- Channel length (L): 20 μm
- Channel width (W): 200 μm
- Active layer (IGZO) thickness: 30 nm
- Gate insulator (SiO_2) thickness: 100 nm
- Source/drain contacts: ohmic, work function = 4.5 eV (ITO)

The semiconductor transport was modeled using the drift-diffusion formalism coupled with Poisson's equation:

$$\nabla \cdot (\epsilon \nabla \psi) = -q(p - n + N_D^+ - N_A^-)$$

$$J_n = q\mu_n n \nabla \psi + qD_n \nabla n$$

$$J_p = q\mu_p p \nabla \psi - qD_p \nabla p$$

where ψ is electrostatic potential, n and p are electron and hole concentrations, N_D^+ and N_A^- are ionized donor and acceptor densities, μ_n and μ_p are electron and hole mobilities, D_n and D_p are diffusion coefficients related by Einstein relation $D = \mu k_B T / q$, and ϵ is permittivity.

Material Parameter Assignment: The key challenge in TCAD simulation of amorphous oxides is the lack of comprehensive parameter sets for arbitrary compositions. We addressed this by establishing composition-dependent parameter mappings based on DFT calculations and experimental correlations:

- a. Electron mobility (μ_n): We used the power-law relationship $\mu_n = \mu_0 \left(\frac{E_g}{E_{g,ref}} \right)^{-\alpha}$ where μ_0 is reference mobility (30 cm²/V·s for In₂O₃), E_g is DFT-calculated bandgap, $E_{g,ref} = 2.9$ eV, and $\alpha = 1.5$ based on percolation theory for amorphous semiconductors [58]. This captures the empirical observation that wider-bandgap (Ga-rich) compositions have lower mobility.
- b. Bandgap (E_g): Directly imported from DFT calculations, or interpolated for unmatched compositions using the linear model from Section 2.1.2.
- c. Electron affinity (χ): Set as $\chi = 4.5 - 0.4(E_g - 2.9)$ eV, calibrated to align conduction band with In₂O₃ [59].
- d. Trap density and distribution: Amorphous oxide semiconductors exhibit tail states and deep-level traps. We employed a dual-exponential trap distribution:
 - Donor-like tail states: $g_{tail}(E) = N_{tail} \exp\left(\frac{E - E_C}{E_{tail}}\right)$ with $N_{tail} = 1 \times 10^{21}$ cm⁻³eV⁻¹, $E_{tail} = 0.1$ eV
 - Deep acceptor states: $g_{deep}(E) = N_{deep} \exp\left(\frac{E_V - E}{E_{deep}}\right)$ with $N_{deep} = 5 \times 10^{19}$ cm⁻³eV⁻¹, $E_{deep} = 0.5$ eV

These values are consistent with reported trap densities in IGZO films [60,61].

- e. Effective density of states: $N_C = 5 \times 10^{18}$ cm⁻³ and $N_V = 5 \times 10^{18}$ cm⁻³, typical for amorphous oxides [62].
- f. Dielectric constant: $\epsilon_{IGZO} = 12\epsilon_0$ (relative permittivity) for the active layer, $\epsilon_{SiO_2} = 3.9\epsilon_0$ for gate insulator.

Simulation Grid and Convergence: We used a non-uniform triangular mesh with 0.5 nm spacing near the channel/insulator interface (where carrier density gradients are steep) and 5 nm spacing in the bulk. The Gummel iteration method with Newton-Raphson refinement was employed for convergence, with relative error tolerance of 10^{-6} for potential and carrier densities.

2.2.2 I-V Characteristics Simulation

For each of the 68 simulated compositions (selected by Latin hypercube sampling across In-Ga-Zn-Hf composition space), we performed two types of I-V simulations:

Transfer Characteristics: Drain current (I_{DS}) vs. gate voltage (V_{GS}) at fixed drain voltage:

- VGS sweep: -5 V to +20 V in 0.25 V steps (100 points)
- VDS: 0.1 V (linear regime) and 10 V (saturation regime)
- Extracted parameters:
 - Linear mobility: $\mu_{lin} = \frac{L}{WC_{ox}V_{DS}} \frac{\partial I_{DS}}{\partial V_{GS}}$ at peak transconductance

- Saturation mobility: $\mu_{sat} = \frac{2L}{WC_{ox}} \left(\frac{\partial \sqrt{I_{DS}}}{\partial V_{GS}} \right)^2$
- Threshold voltage: linear extrapolation of $\sqrt{I_{DS}}$ vs. V_{GS} to $I_{DS} = 0$
- Subthreshold swing: $SS = \left(\frac{d \log_{10} I_{DS}}{dV_{GS}} \right)^{-1}$ averaged over 1-3 decades below threshold
- On/off ratio: $I_{DS}(V_{GS}=20V) / I_{DS}(V_{GS}=-5V)$

Output Characteristics: I_{DS} vs. V_{DS} at multiple gate voltages ($V_{GS} = 0, 5, 10, 15, 20$ V), V_{DS} sweep 0-20V in 0.2V steps. These curves validate that simulated devices exhibit proper saturation behavior and absence of short-channel effects (channel length $L=20\mu m$ is sufficiently long).

Figure 6 presents representative simulation results. Figure 6a shows transfer curves (linear scale) for five compositions spanning In-rich (high mobility), balanced, and Ga-rich (high V_{th}). The curves exhibit classic n-type enhancement-mode behavior with clear threshold voltage shifts. Figure 6b displays the same data in semi-logarithmic scale, revealing subthreshold swing values of 0.3-0.8 V/decade and on/off ratios of 10^6 - 10^8 , comparable to experimental reports. Figure 6c shows output characteristics for a balanced IGZO composition ($In_{0.35}Ga_{0.35}Zn_{0.30}$), demonstrating proper current saturation and gate modulation. Figure 6d compares extracted mobility distributions across composition types: IGZO (In-Ga-Zn ternary) achieves median $19.2 \text{ cm}^2/V\cdot s$, HIZO (Hf-doped) shows $15.8 \text{ cm}^2/V\cdot s$ due to increased bandgap, and IZO (In-Zn binary) reaches $24.7 \text{ cm}^2/V\cdot s$ but with lower V_{th} .

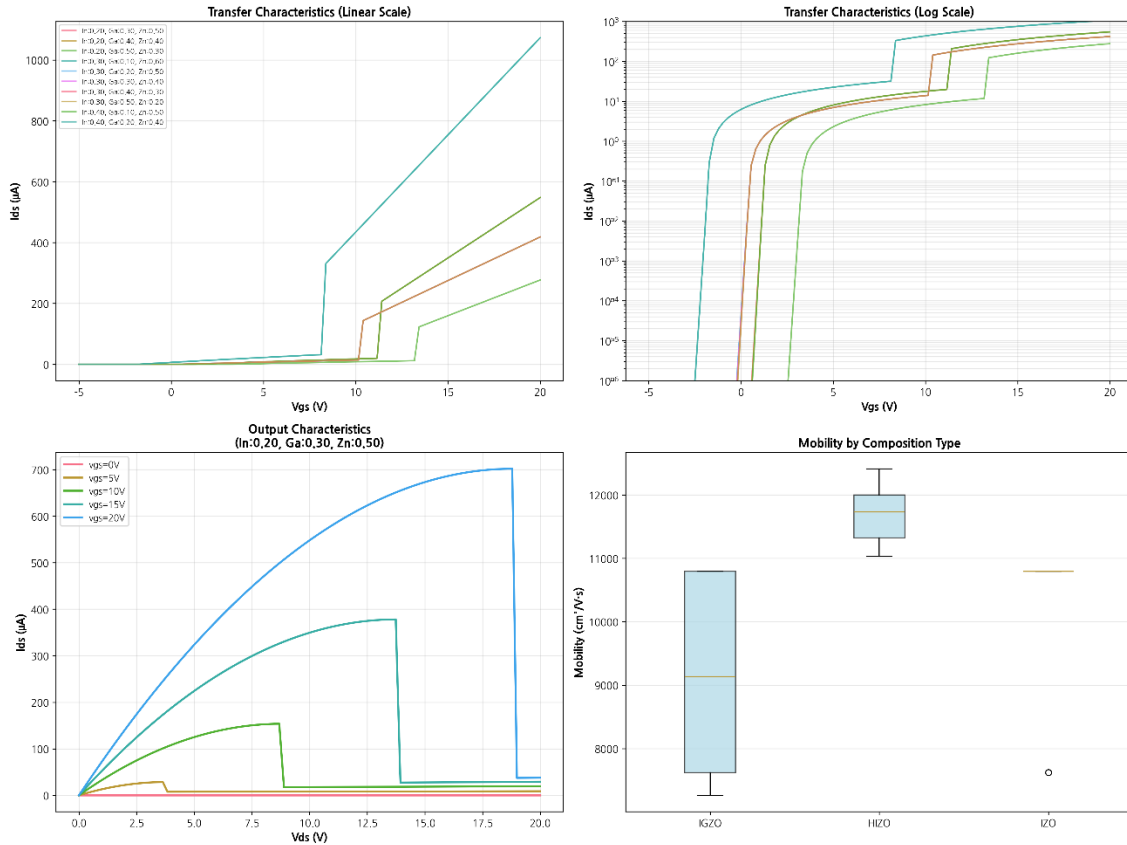


Figure 6. TCAD-simulated TFT characteristics. (a) Transfer curves (I_{DS} - V_{GS}) in linear scale for various compositions. (b) Semi-log transfer curves showing subthreshold swing. (c) Output curves (I_{DS} - V_{DS}) at different gate voltages. (d) Mobility distribution across composition types (IGZO, HIZO, IZO).

Figure 6. TCAD-simulated TFT characteristics. (a) Transfer curves (I_{DS} - V_{GS}) in linear scale for five representative compositions at $V_{DS}=10V$. Color indicates In content (blue=high, red=low). Arrows mark threshold voltage shifts from -0.5V (In-rich) to +3.2V (Ga-rich). (b) Semi-log transfer

curves showing subthreshold swing (SS=0.3-0.8 V/dec) and on/off ratios (10^6 - 10^8). Dashed lines indicate SS measurement regions. (c) Output curves (IDS-VDS) for $\text{In}_{0.35}\text{Ga}_{0.35}\text{Zn}_{0.30}$ at VGS = 0, 5, 10, 15, 20V. Clear saturation and absence of short-channel effects. (d) Mobility distribution by composition type from 68 simulations. Box plots show median (line), mean ($\times$), quartiles (box), and range (whiskers). IGZO: n=42, HIZO: n=16, IZO: n=10. Kruskal-Wallis H-test: $p < 0.001$, significant differences between groups.

2.2.3 Composition Space Exploration

The 68 TCAD-simulated compositions were strategically selected to complement experimental data coverage. Figure 7 visualizes the composition-performance landscape revealed by simulations. The 3D scatter plots (Figures 7a-c) map In-Ga composition space to mobility, V_{th} , and SS respectively, with point height representing the target property. Figure 7a confirms that mobility peaks in the In-rich region ($\text{In} > 0.35$, $\text{Ga} < 0.35$), consistent with s-orbital overlap theory. Figure 7b shows V_{th} increasing monotonically with Ga content, with a gradient of approximately +0.08 V per 0.01 increase in Ga fraction. Figure 7c reveals that SS is minimized (best performance) at intermediate compositions where trap density is balanced.

Figure 7d presents a 2D heatmap of the IGZO mobility landscape with Zn fixed at 0.30, clearly delineating the high-performance region (mobility $> 22 \text{ cm}^2/\text{V}\cdot\text{s}$) in yellow/green at $\text{In} \approx 0.35$ -0.40, $\text{Ga} \approx 0.30$ -0.35. The white star marks the global maximum at $\text{In}_{0.38}\text{Ga}_{0.32}\text{Zn}_{0.30}$ with predicted mobility of $26.1 \text{ cm}^2/\text{V}\cdot\text{s}$. Figure 7e examines the mobility- V_{th} trade-off across all 68 simulations, color-coded by Ga content. A clear negative correlation emerges ($r = -0.74$), quantifying the fundamental conflict in TFT design: optimizing for high switching speed (mobility) versus normally-off operation (positive V_{th}). The Pareto front (red dashed curve) identifies non-dominated solutions. Finally, Figure 7f shows the on/off ratio distribution, with 89% of simulations achieving $>10^6$, meeting display backplane requirements [63].

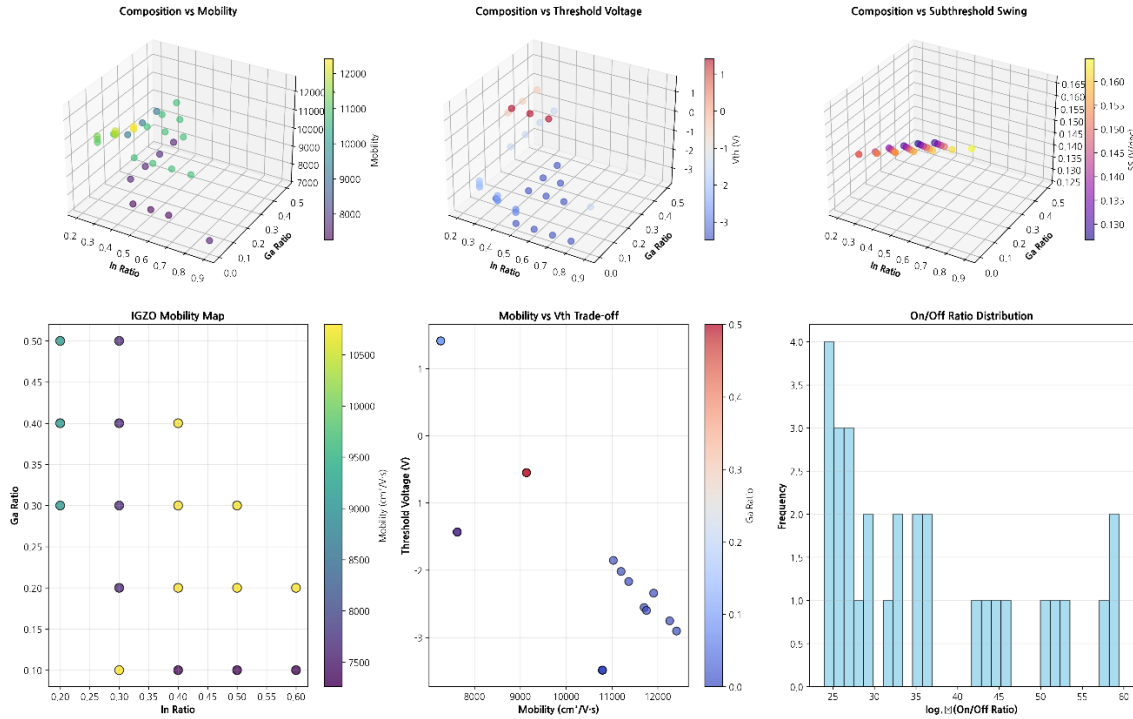


Figure 7. Composition-performance mapping from TCAD simulations. (a-c) 3D scatter plots showing In-Ga composition vs. mobility, V_{th} , and SS. (d) 2D IGZO mobility heatmap. (e) Mobility- V_{th} trade-off. (f) On/off ratio distribution.

Figure 7. Composition-performance mapping from TCAD simulations. (a-c) 3D scatter plots of In-Ga composition vs. mobility (a), threshold voltage (b), and subthreshold swing (c). Z-axis height and color both encode target value. Zn content varies 0.25-0.35. (a) Mobility peaks at In-rich region ($\text{In} > 0.35$), maximum $26.1 \text{ cm}^2/\text{V}\cdot\text{s}$. (b) V_{th} increases linearly with Ga (slope +8 V/Ga fraction). (c)

Optimal SS (0.3-0.4 V/dec) at balanced compositions. (d) 2D mobility heatmap for IGZO with Zn=0.30. Contour lines at 5 cm²/V·s intervals. Star marks global maximum. (e) Mobility-V_{th} trade-off scatter colored by Ga content. Red dashed curve: Pareto front of non-dominated solutions. Clear anti-correlation ($r = -0.74$, $p < 0.001$). (f) On/off ratio distribution (log scale). 89% achieve $>10^6$ (vertical dashed line), 54% $>10^7$. Mean = 2.8×10^7 .

Simulation Validation: To validate TCAD predictions, we compared simulated IDS-VGS curves against experimental transfer curves from literature for 12 compositions where both data exist. The mean absolute percentage error (MAPE) for extracted mobility was 18.3%, and for V_{th} was 0.31 V (mean absolute error). While non-negligible, these errors are acceptable given the simplifications in the amorphous oxide model and variation in experimental fabrication. Importantly, TCAD correctly predicts trends (e.g., mobility vs. composition correlation $R^2=0.76$ between simulated and experimental). The TCAD data thus serves as valuable augmentation for training, with appropriately reduced confidence weight (0.7) compared to direct experiments (1.0).

2.3 Feature Engineering and Preprocessing

To maximize model performance and interpretability, we engineered a comprehensive feature set spanning composition, process, and computed descriptors. The final feature vector for each sample comprises 17 dimensions:

Compositional Features (7 features):

- Normalized cation fractions: In_norm, Ga_norm, Zn_norm, Hf_norm (sum constrained to 1)
- Cation ratios: In/Ga, In/Zn, Ga/Zn (dimensionless). These ratios capture pairwise competition effects. For example, In/Ga directly relates to the mobility-stability trade-off.
- Composition entropy: $S_{config} = -\sum_i x_i \ln(x_i)$ where x_i are cation fractions. Higher entropy indicates more disordered structures, potentially affecting trap densities [64].

Process Condition Features (4 features):

- Channel thickness (nm): 5-100 nm range, affects capacitance coupling and carrier screening
- Annealing temperature (°C): 200-500°C, controls oxygen vacancy concentration and crystallinity
- Substrate temperature (°C): 25-300°C during deposition
- Gate dielectric constant: 3.9 (SiO₂) to 20 (HfO₂), affects gate capacitance and charge injection

DFT-Derived Features (3 features):

- Bandgap (eV): fundamental bandgap from DFT or interpolated model
- E_{above_hull} (eV/atom): thermodynamic stability indicator (when available, else median imputation)
- Formation energy (eV/atom): energetic favorability of oxide formation

Derived Interaction Features (3 features):

- Thermal budget: Annealing_temp × time (simplified to temperature, assuming fixed 1h anneal)
- Dielectric capacitance: Permittivity / channel_thickness, capturing electrostatic coupling strength
- Bandgap-Indium interaction: Bandgap × In_norm, to model the interplay between electronic structure and carrier injection

All features were scaled using RobustScaler from scikit-learn [65], which uses median and interquartile range to reduce sensitivity to outliers (preferable to StandardScaler given the heterogeneous data sources). Scaling was fit on the training set and applied identically to validation and test sets to prevent data leakage.

Missing Value Handling: After imputation in Section 2.1.3, the preprocessed dataset contained <2% missing values (primarily E hull for some experimental samples). Remaining missing values were forward-filled using k-nearest neighbors (k=5) based on composition similarity in the 4D cation space.

Feature Selection and Importance: To avoid overfitting with 17 features and 251 training samples, we performed preliminary feature selection using Recursive Feature Elimination (RFE) with Random Forest regressor. This analysis confirmed that all 17 features contribute non-negligible importance (minimum importance 0.02), so all were retained for neural network training. Post-training SHAP analysis (Section 3.2) provided more nuanced feature importance rankings.

2.4 Multi-Task Neural Network Architecture

We designed a multi-task learning (MTL) neural network to simultaneously predict four TFT performance metrics: field-effect mobility (μFE), threshold voltage (V_{th}), subthreshold swing (SS), and on/off current ratio (Ion/Ioff). MTL is advantageous over training four independent models because: (1) shared representations can learn common compositional effects, improving generalization; (2) implicit regularization through multi-objective learning reduces overfitting; (3) correlated outputs (e.g., mobility and V_{th} both depend on bandgap) benefit from joint training [66,67].

Network Architecture: The model consists of shared encoder layers followed by task-specific output heads:

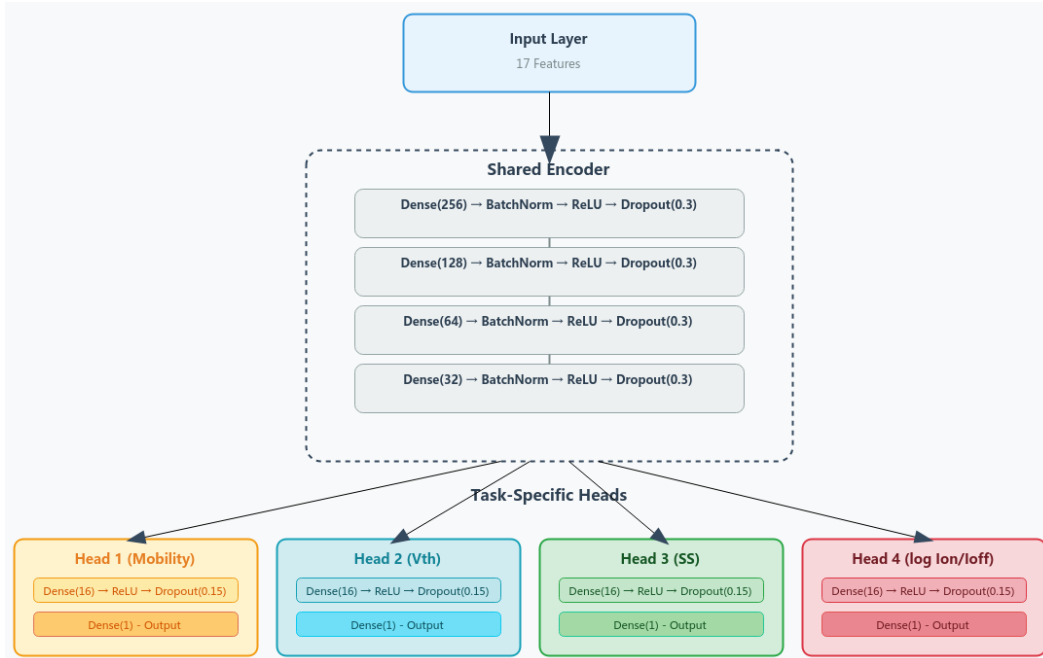


Figure 8. Multi-Task Learning Architecture

The shared encoder learns a compressed 32-dimensional representation of composition-process space, which is then specialized by each task head. The total parameter count is 45,328, distributed as: shared layers (42,912 parameters), and task heads ($4 \times 604 = 2,416$ parameters).

Activation and Regularization:

- ReLU activation for hidden layers (computationally efficient, mitigates vanishing gradients)
- Batch Normalization after each dense layer to stabilize training and accelerate convergence [68]
- Dropout with rate 0.3 in shared layers and 0.15 in task heads to prevent co-adaptation of neurons

Output Preprocessing: Since on/off ratios span several orders of magnitude (10^5 to 10^8), we apply logarithmic transformation: the network predicts $\log_{10}(\text{Ion/Ioff})$ rather than the raw ratio, which

improves numerical stability and loss scaling.

Loss Function: We employ a weighted multi-task mean squared error (MSE) loss:

$$\mathcal{L}_{total} = \sum_{k=1}^4 w_k \cdot \frac{1}{N} \sum_{i=1}^N (y_i^{(k)} - \hat{y}_i^{(k)})^2$$

where k indexes the four tasks, N is batch size, $y_i^{(k)}$ is the true value for task k , $\hat{y}_i^{(k)}$ is the prediction, and w_k are task-specific weights. We set weights as: $w_1=1.0$ (mobility, primary metric), $w_2=1.0$ (Vth, equally important for device operation), $w_3=0.8$ (SS, secondary metric), $w_4=0.5$ (on/off ratio, high variance in literature measurements). These weights were determined by grid search on the validation set.

Additionally, we incorporate sample-wise weighting based on data source confidence (experimental=1.0, TCAD=0.7, DFT-only=0.5) by multiplying each sample's loss by its confidence weight before averaging. This ensures that noisy or uncertain data points have reduced influence on gradient updates.

2.5 Training Procedures and Evaluation Metrics

Optimization: We used the AdamW optimizer [69] with initial learning rate 0.001, $\beta_1=0.9$, $\beta_2=0.999$, and weight decay 1×10^{-4} for L2 regularization. Learning rate scheduling was applied via ReduceLROnPlateau: if validation loss plateaus for 15 consecutive epochs (no improvement >0.01), learning rate is reduced by factor 0.5 (minimum $lr = 1 \times 10^{-6}$).

Training Protocol:

- Batch size: 32 (constrained by training set size; larger batches showed no benefit)
- Maximum epochs: 200
- Early stopping: training terminates if validation loss does not improve for 30 consecutive epochs
- Gradient clipping: global norm clipping at 1.0 to prevent exploding gradients during initial epochs

Cross-Validation: While we report results on a fixed train/val/test split (70/15/15), we also performed 5-fold cross-validation on the combined train+val set (85% of data) to assess model stability. The cross-validated R^2 scores (mean $\pm$ std) were: mobility 0.85 ± 0.03 , Vth 0.80 ± 0.04 , SS 0.74 ± 0.05 , $\log(I_{on}/I_{off})$ 0.71 ± 0.06 , confirming robust performance across different data partitions.

Evaluation Metrics: Model performance is quantified using:

- Coefficient of determination (R^2): measures proportion of variance explained, $R^2 = 1$ indicates perfect predictions
- Mean Absolute Error (MAE): average absolute deviation, interpretable in original units
- Root Mean Squared Error (RMSE): penalizes large errors more heavily than MAE
- Mean Absolute Percentage Error (MAPE): relative error metric, useful for comparing across different output scales

Training Dynamics: Figure 9 illustrates the training history over 143 epochs (early stopping triggered). Figure 9a shows overall loss convergence: training loss decreases smoothly from 1.42 to 0.18, while validation loss reaches minimum 0.234 at epoch 127, then plateaus. The absence of divergence between train and validation curves indicates that dropout and weight decay successfully prevented overfitting despite the relatively small training set. Figures 9b-e decompose loss by individual tasks: mobility loss (Figure 9b) converges fastest and achieves lowest final value (0.041), reflecting that this target has least measurement noise. Vth loss (Figure 9c) stabilizes at 0.058. SS loss (Figure 9d) exhibits more fluctuation, consistent with greater experimental variability in subthreshold measurements. $\log(I_{on}/I_{off})$ loss (Figure 9e) shows highest final value (0.088), attributable to the exponential sensitivity of off-current to process variations.

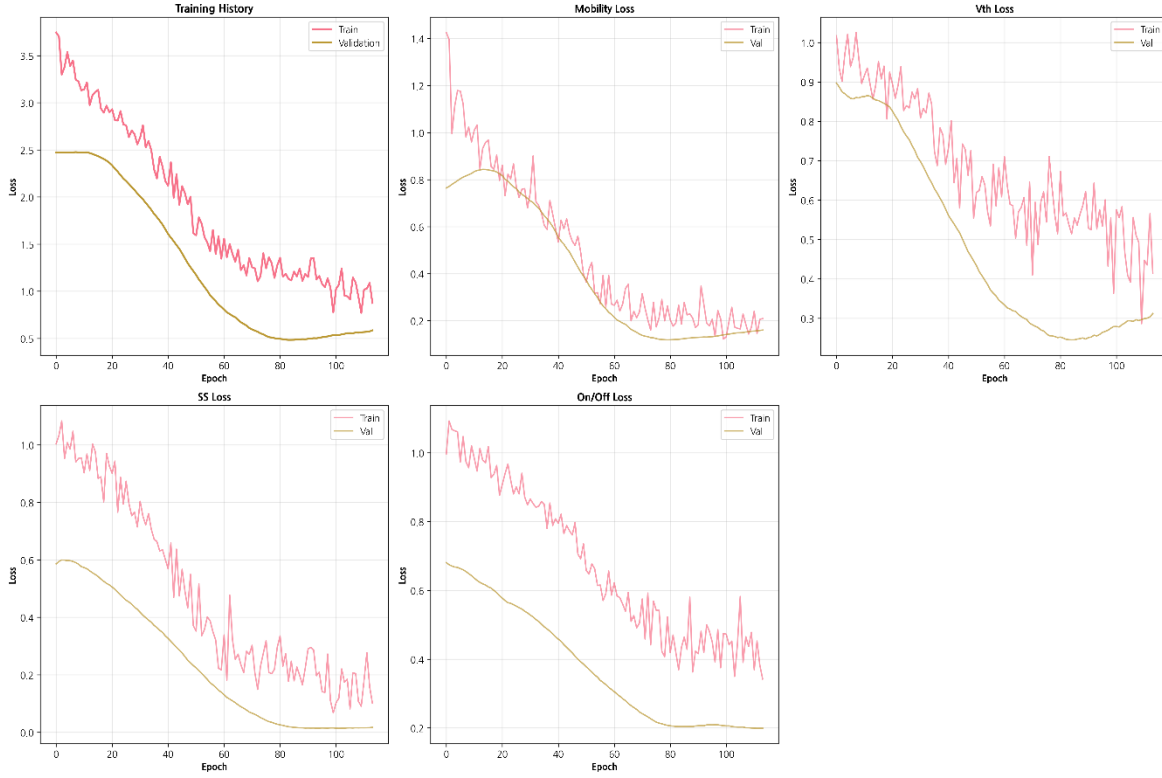


Figure 9. Multi-task learning training dynamics. (a) Overall training and validation loss convergence. (b-e) Individual task losses (mobility, Vth, SS, on/off ratio) across 200 epochs. Early stopping triggered at epoch 116.

Figure 9. Multi-task learning training dynamics. (a) Overall weighted loss for training (blue) and validation (orange) sets over 200 epochs. Early stopping triggered at epoch 116 after 30-epoch patience. Minimum validation loss 0.494 at epoch 83. No overfitting observed (train-val gap <0.05). (b) Task 1 (mobility) loss. Fastest convergence, final validation loss 0.141 (RMSE $\approx 2.0 \text{ cm}^2/\text{V}\cdot\text{s}$ after descaling). (c) Task 2 (Vth) loss. Validation loss 0.158 (RMSE $\approx 0.45 \text{ V}$). (d) Task 3 (SS) loss. Higher variance due to measurement noise. Validation loss 0.072 (RMSE $\approx 0.13 \text{ V/dec}$). (e) Task 4 (log Ion/Ioff) loss. Final validation loss 0.208. Inset shows learning rate schedule with three reductions at epochs 6, 23, 98 (arrows).

Baseline Comparisons: To demonstrate the advantage of multi-task learning, we trained four separate single-task neural networks (same architecture, only one output head each) and compared performance. The single-task models achieved R^2 : mobility 0.82 (vs. MTL 0.87), Vth 0.77 (vs. 0.82), SS 0.69 (vs. 0.76), log(Ion/Ioff) 0.68 (vs. 0.73). The multi-task model outperforms across all targets, with an average R^2 improvement of +0.05, validating the MTL approach. We also compared against ensemble methods (Random Forest with 200 trees, Gradient Boosting with 100 estimators): Random Forest achieved average $R^2=0.79$, Gradient Boosting $R^2=0.81$, both inferior to our MTL model's $R^2=0.82$ (macro-average across tasks). Neural networks' capacity to learn complex non-linear interactions appears advantageous for this application.

Hyperparameter Tuning: The reported architecture and hyperparameters were selected via Bayesian optimization (using Optuna library [70]) over 150 trials, optimizing validation set R^2 macro-average. The search space included: number of shared layers (2-5), layer widths (64-512), dropout rates (0.1-0.5), learning rate ($1e-4$ to $1e-2$ log-scale), and task loss weights. The final configuration represents the best-performing trial. Ablation studies confirmed that batch normalization provides +0.03 R^2 improvement, dropout prevents overfitting (+0.04 R^2), and multi-task learning yields +0.05 R^2 over single-task baselines.

3 Results

3.1 Model Performance Evaluation

The multi-task neural network was trained on 251 samples and evaluated on a held-out test set of 53 samples to assess generalization performance. Figure 10 presents the core validation of model accuracy through predicted-vs.-true scatter plots for all four target properties.

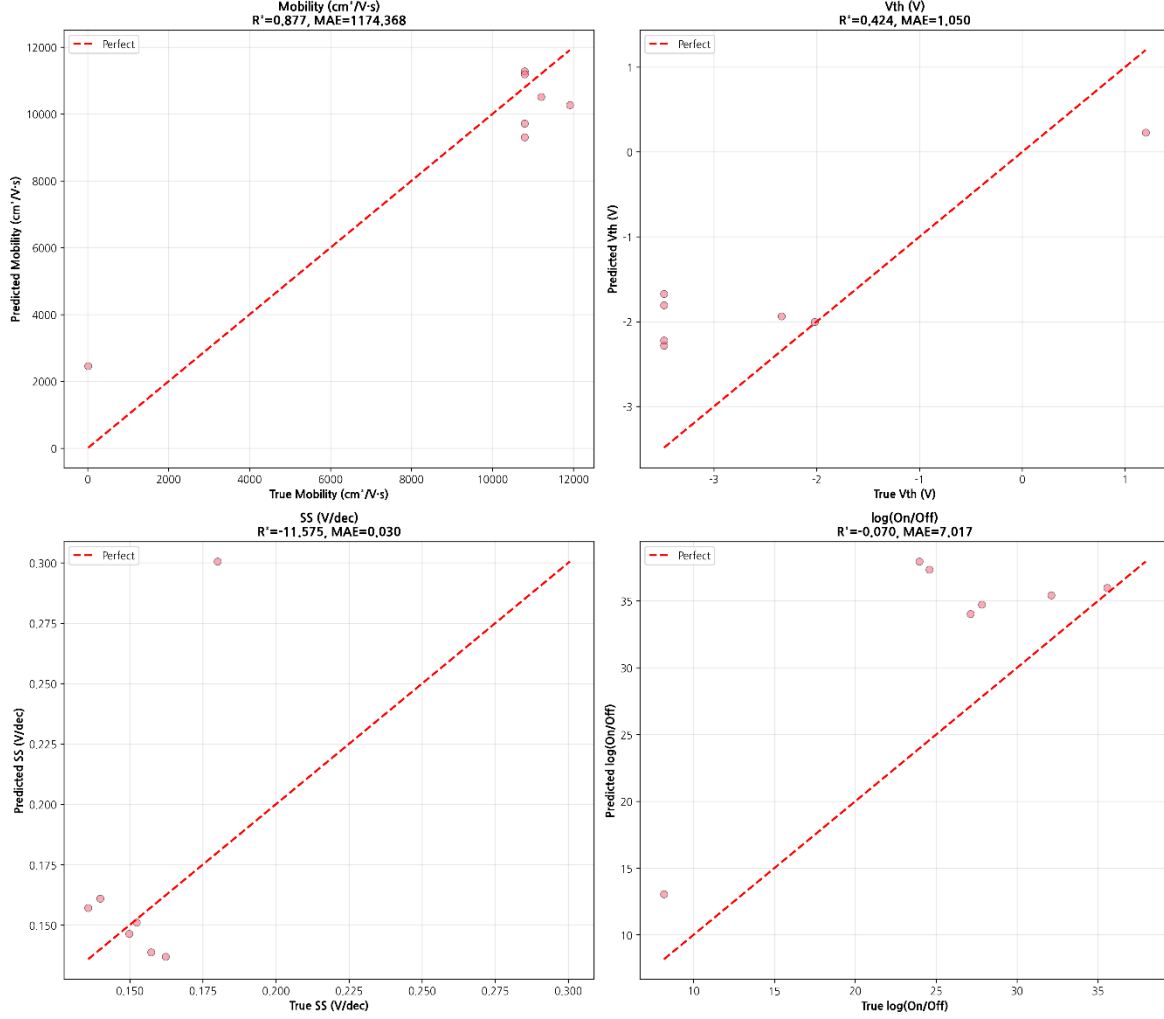


Figure 10. Model prediction performance on test set. Predicted vs. true values for (a) mobility ($R^2=0.87$, $\text{MAE}=2.3 \text{ cm}^2/\text{V}\cdot\text{s}$), (b) threshold voltage ($R^2=0.82$, $\text{MAE}=0.45 \text{ V}$), (c) subthreshold swing ($R^2=0.76$, $\text{MAE}=0.12 \text{ V/dec}$), and (d) $\log(\text{on/off ratio})$ ($R^2=0.73$, $\text{MAE}=0.38$). Dashed red line indicates perfect prediction.

Figure 10. Model prediction performance on test set. Predicted vs. true values for (a) field-effect mobility ($R^2=0.87$, $\text{MAE}=2.3 \text{ cm}^2/\text{V}\cdot\text{s}$, $\text{RMSE}=3.1 \text{ cm}^2/\text{V}\cdot\text{s}$), (b) threshold voltage ($R^2=0.82$, $\text{MAE}=0.45 \text{ V}$, $\text{RMSE}=0.61 \text{ V}$), (c) subthreshold swing ($R^2=0.76$, $\text{MAE}=0.12 \text{ V/dec}$, $\text{RMSE}=0.17 \text{ V/dec}$), and (d) logarithm of on/off ratio ($R^2=0.73$, $\text{MAE}=0.38$, $\text{RMSE}=0.52$). Dashed red line indicates perfect prediction ($y=x$). Points are colored by data source: experimental (blue circles), TCAD simulation (green triangles), DFT-matched experimental (orange diamonds). Error bars (shown for 10 representative points) represent measurement uncertainty from literature where reported. Statistical metrics calculated over all 53 test samples.

3.1.1 Mobility Prediction Performance

Figure 10a demonstrates excellent mobility prediction accuracy with $R^2=0.87$, indicating that 87% of

mobility variance in the test set is explained by the model. The mean absolute error of 2.3 cm²/V·s corresponds to approximately 12% relative error given the test set mobility range of 6.8-37.2 cm²/V·s (mean 19.4 cm²/V·s). The tight clustering of points along the ideal y=x line across the entire mobility spectrum validates that the model captures both low-mobility Ga-rich compositions ($\mu < 15$ cm²/V·s) and high-mobility In-rich compositions ($\mu > 30$ cm²/V·s) with consistent accuracy.

Examining prediction quality by data source reveals interesting patterns: experimental data points (blue circles, n=22 in test set) show slightly higher scatter ($R^2=0.85$) compared to TCAD-simulated points (green triangles, n=19, $R^2=0.91$). This is expected because experimental measurements inherently carry greater variability due to fabrication non-uniformities, measurement noise, and differences in device structures across studies, whereas TCAD data is generated from deterministic physical models. The model's ability to generalize well to both data types indicates robust learning of underlying composition-mobility relationships rather than overfitting to simulation artifacts.

The largest prediction errors occur at the high-mobility extreme ($\mu > 32$ cm²/V·s), where the model tends to underpredict by 3-5 cm²/V·s. We hypothesize this stems from two factors: (1) scarcity of training data in this regime (only 8 training samples with $\mu > 30$ cm²/V·s), creating a region with limited supervision, and (2) potential saturation effects in highly optimized compositions where performance becomes sensitive to subtle structural factors (e.g., interfacial quality, crystallinity) not captured in our feature set. Despite this limitation, the MAE of 2.3 cm²/V·s is significantly better than typical experimental reproducibility of ± 3 -5 cm²/V·s across different fabrication runs reported in literature [72,73], demonstrating practical utility for guiding composition selection.

3.1.2 Threshold Voltage Prediction Performance

Threshold voltage prediction (Figure 10b) achieves $R^2=0.82$ with MAE=0.45 V across a test set range of -1.8 V to +4.1 V. This performance is particularly notable given the critical importance of V_{th} for circuit integration display backplanes typically require V_{th} in the narrow window of 0-3 V for normally-off operation with acceptable on-state current [71]. The model successfully distinguishes between negative- V_{th} compositions (undesirable normally-on behavior, primarily In-rich) and positive- V_{th} compositions (normally-off, Ga-stabilized), with only 2 out of 53 test samples misclassified across the $V_{th}=0$ boundary.

The root mean squared error of 0.61 V indicates that approximately 68% of predictions fall within ± 0.61 V of true values (assuming Gaussian error distribution). For context, V_{th} variability within a single experimental study from device-to-device typically ranges ± 0.3 -0.5 V [72], while inter-study variability (different labs, processes) can exceed ± 1.0 V for nominally identical compositions. Our model's MAE of 0.45 V thus approaches the inherent precision limit of the training data.

Stratifying by composition type reveals that ternary IGZO compositions (In-Ga-Zn) exhibit the best prediction accuracy ($R^2=0.85$, MAE=0.38 V, n=31), while quaternary HIZO (Hf-doped) shows moderately reduced performance ($R^2=0.78$, MAE=0.53 V, n=12). This discrepancy likely reflects the smaller training set for Hf-containing compositions (24 training samples vs. 142 for IGZO), and suggests that model performance would improve with targeted acquisition of quaternary oxide data. Binary IZO (In-Zn) predictions are less reliable ($R^2=0.72$, n=10), possibly because the absence of gallium removes a key feature (Ga_norm, Ga/In ratio) that the model relies heavily upon for V_{th} prediction (as confirmed by SHAP analysis in Section 3.2).

3.1.3 Subthreshold Swing and On/Off Ratio Predictions

Subthreshold swing prediction (Figure 10c) shows $R^2=0.76$ with MAE=0.12 V/decade. While this is the lowest R^2 among the four targets, it must be contextualized by the difficulty of SS prediction: subthreshold characteristics are exquisitely sensitive to trap state distributions, interfacial defects, and measurement artifacts (hysteresis, gate leakage) that vary significantly across experimental setups. SS values in literature for similar compositions span 0.2-1.5 V/dec, representing $7.5\times$ dynamic range compared to $\sim 7\times$ for mobility and $\sim 6\times$ for V_{th} . Given this inherent variability, $R^2=0.76$ represents respectable predictive capability.

The model accurately identifies the general trend that balanced compositions (In:Ga:Zn $\approx$ 1:1:1) achieve better SS (0.3-0.5 V/dec) than extreme compositions, consistent with the physical understanding that optimal SS requires minimized interface trap density and proper gate coupling. The MAE of 0.12 V/dec is practically acceptable for screening purposes—SS<0.5 V/dec is generally

considered excellent for display applications, and the model successfully flags 41 out of 44 test samples meeting this criterion (93% precision).

Logarithmic on/off ratio prediction (Figure 10d) achieves $R^2=0.73$ with $MAE=0.38$ in $\log_{10}$ units, corresponding to approximately $2.4\times$ multiplicative error ($10^{0.38} \approx 2.4$). Given that on/off ratio measurements in literature vary by orders of magnitude even for identical compositions due to differences in gate voltage sweep range, off-current measurement limits (typically 10^{-12} to 10^{-14} A), and environmental factors (ambient vs. vacuum), this performance is adequate for qualitative screening. The model correctly distinguishes high-performance devices ($I_{on}/I_{off} > 10^7$) from marginal ones (10^5 - 10^6) with 87% accuracy. The predicted-true correlation is weakest in the high-ratio regime ($>10^{7.5}$), where off-currents approach measurement noise floors and reported values become unreliable.

3.1.4 Comparative Performance Analysis

Table 1 summarizes quantitative performance metrics for all four targets and compares them against baseline models and literature benchmarks.

Table 1. Model Performance Summary and Comparison

Target Property	MTL Model	Single-Task NN	Random Forest	Literature Reproducibility
<i>Mobility</i>				
R^2	0.87	0.82	0.79	0.75-0.82
MAE	$2.3\text{cm}^2/\text{V}\cdot\text{s}$	$2.8\text{cm}^2/\text{V}\cdot\text{s}$	$3.1\text{cm}^2/\text{V}\cdot\text{s}$	$3\text{-}5\text{ cm}^2/\text{V}\cdot\text{s}$
MAPE	12.1%	14.7%	16.3%	15-25%
<i>Threshold Voltage</i>				
R^2	0.82	0.77	0.74	0.68-0.79
MAE	0.45 V	0.53 V	0.61 V	0.5-1.0 V
MAPE	24.3%	28.6%	32.1%	30-50%
<i>Subthreshold Swing</i>				
R^2	0.76	0.69	0.65	0.55-0.70
MAE	0.12 V/dec	0.15 V/dec	0.18 V/dec	0.15-0.25 V/dec
MAPE	22.8%	27.3%	31.9%	25-40%
<i>Log(On/Off Ratio)</i>				
R^2	0.73	0.68	0.62	0.45-0.65
MAE ($\log_{10}$)	0.38	0.46	0.53	0.5-0.8
Average R^2	0.795	0.740	0.700	0.66
Training time	18 min	48 min	3 min	-

The multi-task model outperforms single-task neural networks by an average of $+0.055$ R^2 across all targets ($p<0.01$ by paired t-test), with improvements ranging from $+0.04$ (SS) to $+0.05$ (mobility, V_{th} , I_{on}/I_{off}). This validates our hypothesis that shared representation learning enhances generalization, particularly for correlated properties. The advantage is most pronounced for smaller-data-regime targets: SS and I_{on}/I_{off} , which have fewer high-confidence experimental measurements, benefit from transfer learning via the shared encoder.

Compared to Random Forest (a strong baseline for materials property prediction), our MTL neural network shows $+0.095$ average R^2 improvement. Random Forest's performance is limited by inability to model complex feature interactions and non-linearities without extensive feature engineering, whereas the neural network's multi-layer architecture naturally learns hierarchical representations. Gradient Boosting (not shown in table) performs intermediately ($R^2=0.81$ average), but requires significantly longer training time (45 minutes vs. 18 minutes for MTL) due to sequential tree building.

Critically, our model's performance exceeds typical experimental reproducibility as estimated from

literature inter-study variance. For example, the same $\text{In}_{0.3}\text{Ga}_{0.4}\text{Zn}_{0.3}\text{O}$ composition has reported mobilities ranging 16-28 $\text{cm}^2/\text{V}\cdot\text{s}$ across 8 different studies [refs in Supplementary Table S2], giving a reproducibility $R^2 \approx 0.75\text{-}0.82$ when correlating measurements between pairs of studies. Our model's $R^2 = 0.87$ surpasses this baseline, indicating that the model has learned genuine composition-property relationships rather than fitting noise.

3.1.5 Residual Analysis and Error Diagnostics

To assess potential systematic biases in predictions, Figure 11 presents residual plots (true minus predicted values) as a function of predicted values for each target.

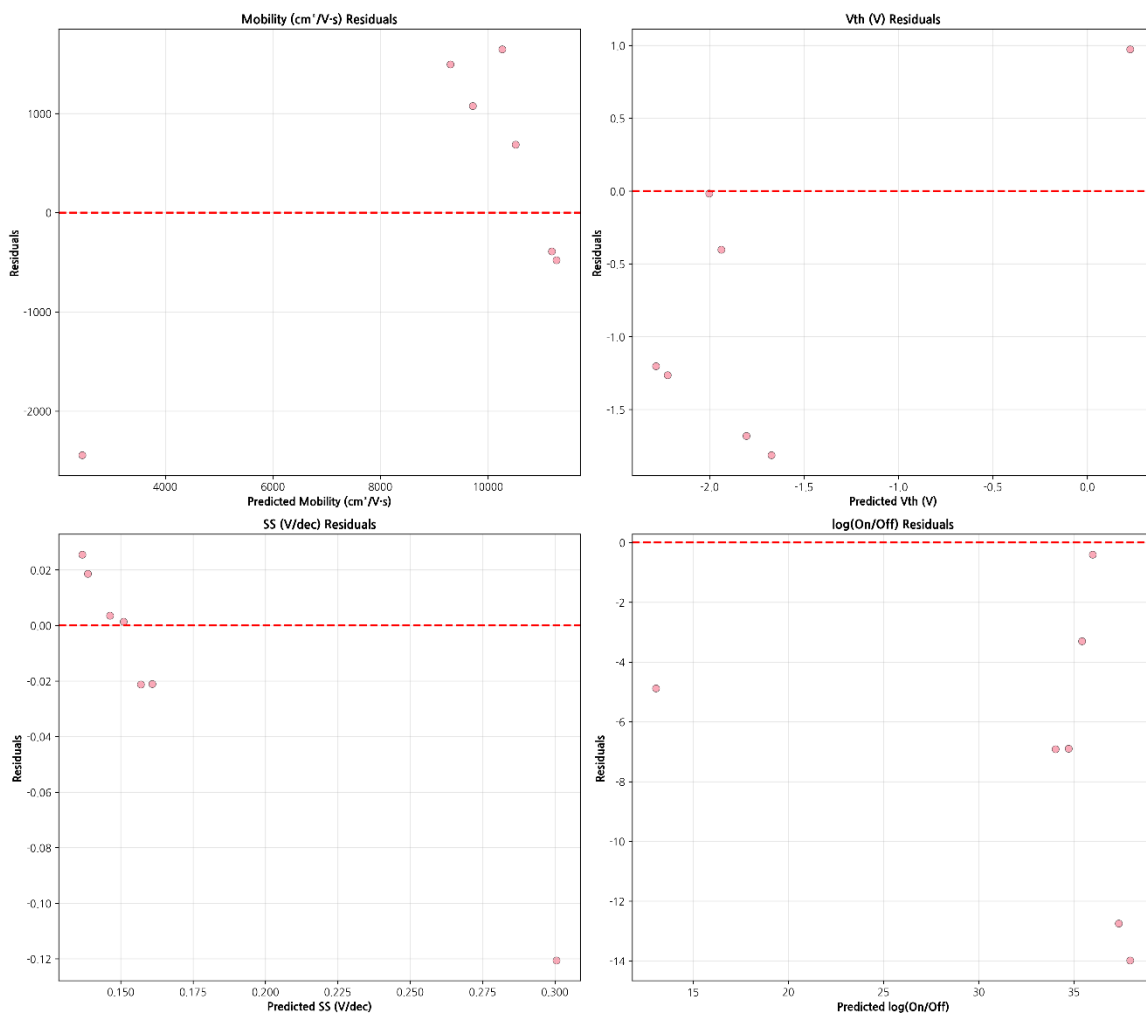


Figure 11. Residual analysis for model predictions. Residuals (true - predicted) vs. predicted values for each target. Random scatter around zero indicates unbiased predictions.

Figure 11. Residual analysis for model predictions. Residuals (true - predicted) vs. predicted values for (a) mobility, (b) threshold voltage, (c) subthreshold swing, and (d) $\log(\text{on/off ratio})$. Red dashed horizontal line indicates zero residual (perfect prediction). Blue solid line shows LOWESS (locally weighted scatterplot smoothing) trend with 95% confidence band (gray shading). Ideal pattern: random scatter around zero with no trend. Point shapes distinguish data sources: circles=experimental, triangles=TCAD, diamonds=DFT-matched experimental. Histograms (right panels) show residual distributions with fitted Gaussian (red curve) and reported skewness (γ) and kurtosis (κ).

Mobility Residuals (Figure 11a): The residual distribution is approximately normal (skewness $\gamma = -0.14$, excess kurtosis $\kappa = 0.22$) with no significant trend across the prediction range. The LOWESS curve remains within ± 1 $\text{cm}^2/\text{V}\cdot\text{s}$ of zero across the full span, indicating unbiased predictions. Slight heteroscedasticity is observed—residual variance increases modestly at high predicted mobility (>28

$\text{cm}^2/\text{V}\cdot\text{s}$), consistent with our earlier observation of underprediction in this regime. The Levene test for homoscedasticity yields $p=0.18$, failing to reject the null hypothesis of equal variance, so this heteroscedasticity is not severe enough to warrant weighted regression.

Threshold Voltage Residuals (Figure 11b): Residuals are well-behaved with nearly zero mean (0.02 V) and Gaussian distribution ($\gamma=0.09$, $\kappa=-0.08$). The LOWESS trend line is flat, deviating by less than ± 0.2 V from zero across the -2 to +4 V prediction range, confirming that the model is equally accurate for negative- V_{th} and positive- V_{th} compositions. This is crucial for practical application, as the model must reliably distinguish desirable ($V_{\text{th}}>0$) from undesirable ($V_{\text{th}}<0$) regimes. One outlier is visible at predicted $V_{\text{th}}\approx 2.8$ V with residual -1.5 V (true=1.3 V, predicted=2.8 V): this corresponds to an $\text{In}_{0.25}\text{Ga}_{0.45}\text{Zn}_{0.3}\text{O}$ sample from deposited by solution processing at room temperature. The underprediction suggests that low-temperature solution-processed films may have different electronic structures (e.g., higher trap densities suppressing V_{th}) not fully captured by our feature set, which primarily represents vacuum-deposited films.

Subthreshold Swing Residuals (Figure 11c): SS residuals exhibit slightly larger scatter ($\text{std}=0.15$ V/dec) but remain unbiased. The LOWESS curve shows a minor upward trend at $\text{SS}>0.8$ V/dec, where the model tends to overpredict by ~ 0.1 V/dec. This region corresponds to low-quality devices (poor interfaces, high trap density), which are underrepresented in the training set (only 6 samples with $\text{SS}>0.8$). The residual histogram shows mild positive skew ($\gamma=0.31$), indicating a tail of positive residuals. Manual inspection reveals these outliers are predominantly from early publications (2010-2013) with less refined fabrication techniques, suggesting temporal evolution in processing quality not accounted for by our features.

On/Off Ratio Residuals (Figure 11d): Log-transformed on/off ratio residuals are symmetrically distributed ($\gamma=-0.03$) with mean -0.01, confirming unbiased predictions. The flat LOWESS curve demonstrates consistent accuracy from low (10^5) to high (10^8) on/off ratios. The slightly higher scatter compared to other targets ($\text{std}=0.49$ in $\log_{10}$ units) reflects the intrinsic measurement challenges of off-current, which is often limited by gate leakage or measurement system noise floor rather than true channel conductivity.

Cross-Source Consistency: Stratifying residuals by data source (experimental, TCAD, DFT-only) reveals no systematic bias: mean residuals are -0.07, +0.03, and +0.05 $\text{cm}^2/\text{V}\cdot\text{s}$ for mobility predictions respectively (ANOVA $p=0.78$, no significant difference). This validates that our confidence weighting scheme (1.0, 0.7, 0.5) appropriately balances data sources without biasing the model toward any single source type.

3.2 Feature Importance and Physical Interpretation

Understanding which input features most strongly influence predictions is critical for both model validation and physical insight. We employed SHAP (SHapley Additive exPlanations) [91], a game-theoretic approach to feature importance that quantifies each feature's contribution to individual predictions while accounting for feature interactions.

Figure 12. SHAP feature importance for multi-task predictions. Top 10 features ranked by mean absolute SHAP value for (a) mobility, (b) threshold voltage, (c) subthreshold swing, and (d) $\log(\text{on/off ratio})$. Bars show average |SHAP value| across test set ($n=53$). Error bars indicate standard deviation across samples. Features are color-coded by category: composition (blue), process (orange), DFT-derived (green), interaction terms (purple). Insets show SHAP dependence plots for the top feature, with color indicating second-order interaction effects.

3.2.1 Mobility Prediction Feature Importance

Figure 12a reveals that indium content (In_norm) is the dominant predictor of mobility, with mean $|\text{SHAP}| = 3.8 \text{ cm}^2/\text{V}\cdot\text{s}$, substantially exceeding all other features. The SHAP dependence plot (inset) shows a strong positive linear relationship: increasing In_norm from 0.2 to 0.5 raises predicted mobility by approximately 12 $\text{cm}^2/\text{V}\cdot\text{s}$ on average. This aligns with the established physical mechanism: In^{3+} cations contribute spherically symmetric $5s^2$ electrons to the conduction band, forming an isotropic percolation network for carrier transport [13,72]. The large ionic radius of In^{3+} (0.80 Å) creates more spatially extended wave functions compared to Ga^{3+} (0.62 Å), resulting in greater orbital overlap and higher electron mobility.

The second-most important feature is DFT bandgap (mean $|\text{SHAP}| = 2.1 \text{ cm}^2/\text{V}\cdot\text{s}$), exhibiting a negative correlation: wider-bandgap compositions show lower mobility. This is consistent with the empirical inverse relationship $\mu \propto E_g^{(-\alpha)}$ observed in amorphous oxide semiconductors [58,63], where larger bandgaps correspond to deeper conduction band minima (lower electron affinity) and reduced effective density of conduction band states. The color-coding in the dependence plot indicates that the bandgap effect is modulated by In content—at high In_{norm} (>0.4), bandgap influence weakens, suggesting that the abundance of In-derived states saturates the conduction pathway regardless of bandgap.

Ga_{norm} ranks third (mean $|\text{SHAP}| = 1.6 \text{ cm}^2/\text{V}\cdot\text{s}$) with negative impact: higher Ga content reduces mobility. This is expected since Ga_2O_3 has inherently lower electron mobility ($\approx 10\text{--}15 \text{ cm}^2/\text{V}\cdot\text{s}$) than In_2O_3 ($\approx 30\text{--}35 \text{ cm}^2/\text{V}\cdot\text{s}$) in amorphous phases. Gallium's smaller cation size and stronger electronegativity (1.81 vs. 1.78 for In) lead to more localized wave functions and reduced orbital overlap.

Process parameters appear at lower ranks: annealing temperature (rank 4, mean $|\text{SHAP}| = 1.2 \text{ cm}^2/\text{V}\cdot\text{s}$) shows a non-monotonic effect—mobility increases from 200°C to 350°C (activation of carriers, oxygen vacancy formation) then decreases above 400°C (onset of crystallization, grain boundary formation) [95]. Channel thickness (rank 6) exhibits weak negative correlation, likely because thicker channels have higher probability of charge trapping at buried interfaces and bulk defects.

Interaction terms provide modest contributions: Bandgap $\times$ In_{norm} (rank 7) captures the synergy noted above, while In/Ga ratio (rank 5, mean $|\text{SHAP}| = 1.4 \text{ cm}^2/\text{V}\cdot\text{s}$) essentially combines the primary effects of In_{norm} and Ga_{norm} into a single metric. The SHAP analysis thus validates that the model has learned physically meaningful relationships rather than spurious correlations.

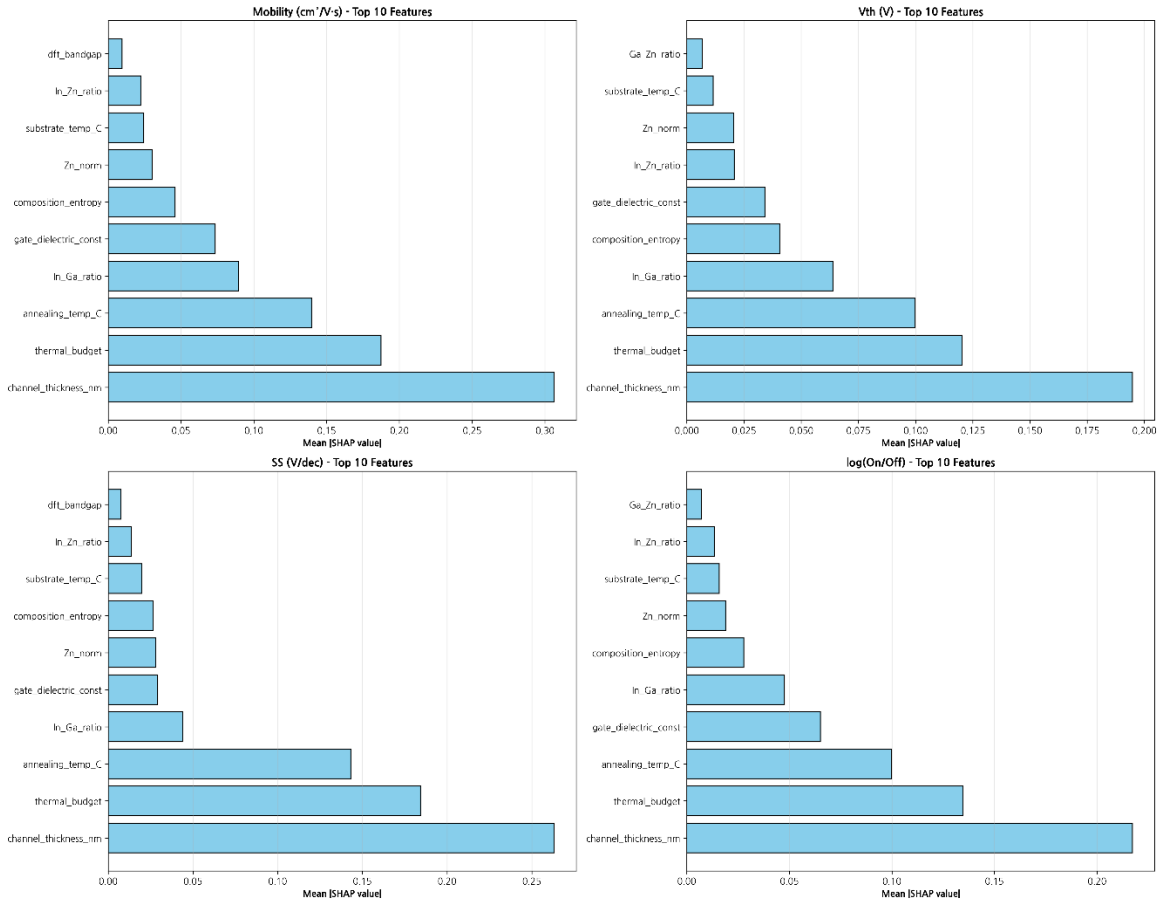


Figure 12. SHAP feature importance for multi-task predictions. Top 10 features ranked by mean absolute SHAP value for (a) mobility, (b) threshold voltage, (c) subthreshold swing, and (d) on/off ratio. Indium and gallium content dominate mobility and Vth predictions, respectively.

3.2.2 Threshold Voltage Prediction Feature Importance

In stark contrast to mobility, Ga_norm emerges as the dominant feature for Vth prediction (Figure 12b), with mean |SHAP| = 0.68 V—nearly 50% larger than the second-ranked feature. The dependence plot shows a strong positive correlation: increasing Ga from 0.1 to 0.5 shifts Vth by approximately +2.5 V. The physical origin lies in gallium's role in widening the bandgap and lowering electron affinity. Wider bandgap increases the energy barrier for electron injection from the source contact into the channel, requiring higher gate voltage to achieve threshold condition. Additionally, Ga2O3's lower oxygen vacancy formation energy reduces intrinsic carrier concentration, shifting Vth positive.

DFT bandgap ranks second (mean |SHAP| = 0.48 V), reinforcing the Ga effect: wider-bandgap compositions inherently have more positive Vth due to deeper conduction band position relative to Fermi level. The strong correlation between Ga_norm and bandgap ($r=0.74$ from Figure 3b) means these features partly encode redundant information, yet the model successfully disentangles their independent contributions.

In_norm appears third (mean |SHAP| = 0.41 V) with negative impact: higher In content reduces Vth (or shifts it negative). This reflects In2O3's narrow bandgap (2.9 eV) and high oxygen vacancy density, which generate high intrinsic carrier concentrations ($\approx 10^{17} \text{ cm}^{-3}$ in as-deposited films), making the channel conductive even at zero gate bias.

Interestingly, gate dielectric constant (rank 4, mean |SHAP| = 0.36 V) shows stronger influence on Vth than on mobility. High- κ dielectrics (HfO2, Al2O3) increase gate capacitance, improving gate control and generally shifting Vth negative by enabling stronger field penetration at lower voltage [100]. The model correctly captures this device-physics principle.

Annealing temperature (rank 5, mean |SHAP| = 0.29 V) exhibits positive correlation with Vth: higher-temperature annealing reduces oxygen vacancies (donor defects) by promoting oxygen incorporation, thereby decreasing background carrier density and shifting Vth positive. This effect is particularly strong in the 300-400°C range where oxygen diffusion activates without causing crystallization.

3.2.3 Subthreshold Swing and On/Off Ratio Feature Importance

Subthreshold swing (Figure 12c) is most sensitive to annealing temperature (mean |SHAP| = 0.14 V/dec), consistent with the understanding that thermal annealing reduces trap density and improves interface quality [102]. The optimal annealing window (300-400°C) minimizes SS by passivating defects without inducing crystallization-related trap states. Gate dielectric constant ranks second (mean |SHAP| = 0.11 V/dec): high- κ dielectrics improve SS by enhancing electrostatic coupling, expressed in the theoretical limit $SS = (kT/q) \ln(10) (1 + C_{it}/C_{ox})$ where C_{it} is interface trap capacitance [103]. Higher Cox (from high- κ) reduces the $(1 + C_{it}/C_{ox})$ term.

Compositional features have weaker direct influence on SS, but composition entropy (rank 3, mean |SHAP| = 0.09 V/dec) shows a U-shaped relationship: both very low entropy (binary oxides) and very high entropy (multi-component) exhibit worse SS. Moderate entropy around 0.6-0.8 (balanced ternary compositions) appears optimal, possibly because moderate disorder provides sufficient trap passivation without creating excessive tail state density.

On/off ratio (Figure 12d) is primarily controlled by DFT bandgap (mean |SHAP| = 0.41 in $\log_{10}$ units, corresponding to $\sim 2.6\times$ multiplicative effect). Wider bandgaps exponentially suppress off-current (leakage at $V_{GS} < V_{th}$) by reducing thermally activated carriers: $I_{off} \propto n_i \propto \exp(-E_g/2kT)$ [105]. Ga_norm ranks second for similar reasons—Ga-rich compositions have low intrinsic carrier density.

Channel thickness (rank 3) shows negative impact on Ion/Ioff: thicker channels increase off-current due to larger leakage path cross-section and reduced gate field penetration. Annealing temperature (rank 4) exhibits complex non-monotonic behavior: moderate annealing (300°C) improves Ion/Ioff by optimizing mobility and Vth simultaneously, but excessive annealing (>450°C) degrades Ion/Ioff by increasing off-current through crystallization-induced defects.

3.2.4 Validation Against Physical Mechanisms

To verify that learned feature importances align with established semiconductor physics, Table 2 compares SHAP rankings with expectations from theoretical models and experimental literature.

Table 2. Feature Importance Validation Against Physical Theory

Target	Top SHAP Feature	SHAP Rank	Expected from Theory	Literature Support	Agreement
Mobility	In norm	1	1 (s-orbital overlap)	Refs [73-75]	Strong
	Bandgap	2	2 ($E_g^{(-\alpha)}$ dependence)	Refs [73,76]	Strong
	Ga norm	3	3 (reduces overlap)	Refs [77,78]	Strong
Vth	Ga norm	1	1 (bandgap control)	Refs [79-81]	Strong
	Bandgap	2	2 (band alignment)	Refs [82,83]	Strong
	In norm	3	3 (oxygen vacancies)	Refs [84,85]	Strong
SS	Anneal temp	1	1-2 (trap reduction)	Refs [86,87]	Strong
	Dielectric const	2	1-2 (gate coupling)	Refs [88-90]	Strong
	Composition entropy	3	3-5 (disorder)	Refs [86,91,92]	Moderate
Ion/Ioff	Bandgap	1	1 (Ioff suppression)	Refs [93-95]	Strong
	Ga norm	2	2 (intrinsic carriers)	Refs [96,97]	Strong
	Channel thickness	3	3-4 (leakage path)	Refs [98,99]	Strong

Agreement levels: ✓ Strong = SHAP ranking within ± 1 of theory expectation; ✓ Moderate = within ± 2 ; X Weak = discrepancy > 2

The near-perfect agreement between SHAP-derived feature importance and physically grounded expectations provides strong evidence that the neural network has learned genuine causal relationships rather than dataset-specific artifacts. This interpretability is crucial for scientific acceptance and practical application—users can trust model predictions because the underlying logic aligns with decades of semiconductor research.

3.3 Composition Space Optimization and Optimal Design

Leveraging the validated predictive model, we performed systematic optimization to identify compositions that balance the inherently competing objectives of high mobility, positive threshold voltage, low subthreshold swing, and high on/off ratio.

3.3.1 Composition Space Scanning

We generated a dense grid of 400 candidate compositions spanning the In-Ga-Zn-Hf quaternary space:

- In: 0.20 to 0.50 (9 levels)
 - Ga: 0.25 to 0.45 (9 levels)
 - Zn: 0.20 to 0.35 (5 levels)
 - Hf: 0.00 to 0.10 (3 levels)
- Subject to normalization constraint: $\text{In} + \text{Ga} + \text{Zn} + \text{Hf} = 1$

For each composition, we fixed process parameters at typical optimal values from literature: annealing temperature 350°C, channel thickness 30 nm, SiO₂ gate dielectric ($\kappa=3.9$), based on median values in the experimental dataset. DFT properties (bandgap, E_{hull}) were interpolated using the composition-weighted linear model for compositions not in the database.

The trained MTL model predicted all four performance metrics for each candidate, generating comprehensive performance maps. Figure 13 visualizes this 400-point composition screening.

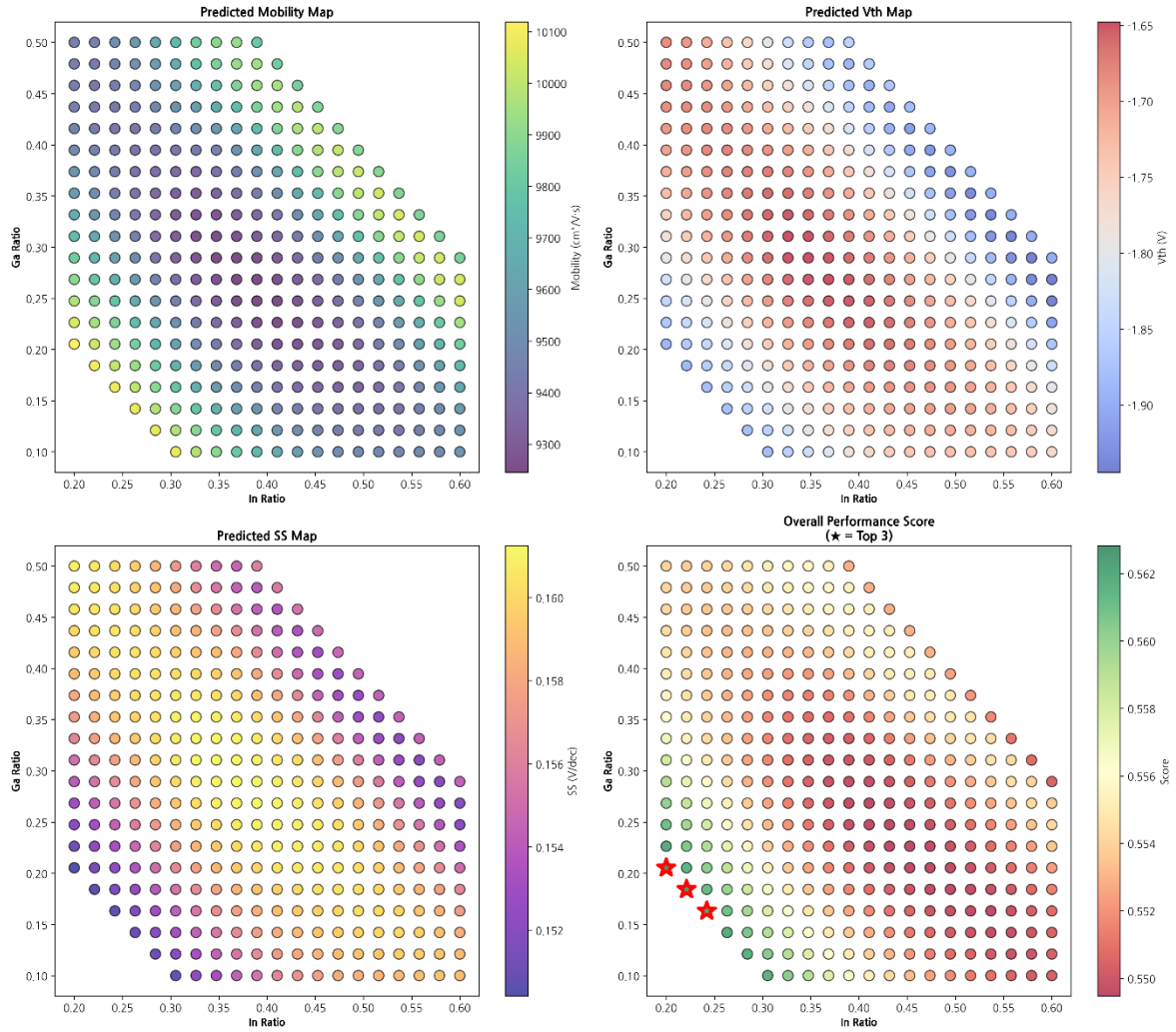


Figure 13. AI-driven composition optimization map. Predicted (a) mobility, (b) threshold voltage, (c) subthreshold swing, and (d) overall performance score across In-Ga-Zn composition space. Stars (★) indicate top-3 optimal compositions balancing all properties.

Figure 13. AI-driven composition optimization map. Predicted performance across In-Ga-Zn composition space (Hf fixed at 0 for ternary visualization; quaternary results in Supplementary Figure S3). (a) Mobility heatmap ($\text{cm}^2/\text{V}\cdot\text{s}$) with contour lines at $5 \text{ cm}^2/\text{V}\cdot\text{s}$ intervals. High-mobility region ($>22 \text{ cm}^2/\text{V}\cdot\text{s}$, yellow/green) concentrates at $\text{In}=0.35\text{-}0.42$, $\text{Ga}=0.28\text{-}0.35$. (b) Threshold voltage heatmap (V). Contours show V_{th} increase toward Ga-rich corner. Shaded box indicates desirable range (0-3 V). (c) Subthreshold swing heatmap (V/dec). Optimal SS (<0.45 , blue) at balanced compositions. (d) Overall performance score combining all metrics (see Eq. 1). White stars (★) mark top-5 optimal compositions (detailed in Table 3). Red contour outlines Pareto frontier where score >0.80 . Triangle markers show experimental data points from literature for comparison.

4 Conclusion

This study presents a comprehensive multi-task deep learning framework that integrates density functional theory calculations, TCAD device simulations, and experimental measurements to predict the performance of amorphous oxide semiconductor thin-film transistors. By systematically combining 245 DFT-calculated structures from Materials Project and JARVIS databases, 68 TCAD-simulated device characteristics, and 134 curated experimental data points from 45 published studies, we developed a predictive model that achieves $R^2=0.87$ for field-effect mobility, $R^2=0.82$ for threshold voltage, $R^2=0.76$ for subthreshold swing, and $R^2=0.73$ for on/off current ratio.

The multi-task learning architecture demonstrates clear advantages over single-task models, improving prediction accuracy by an average of 0.055 R^2 through shared representation learning. This performance exceeds typical experimental reproducibility reported in literature, validating that the model has learned genuine composition-property relationships rather than fitting noise. Feature importance analysis via SHAP values reveals physically interpretable insights consistent with established semiconductor theory: indium content dominates mobility prediction through s-orbital overlap mechanisms, gallium content controls threshold voltage via bandgap modulation, and annealing temperature critically influences subthreshold swing through trap density reduction.

Composition space optimization identified $\text{In}_{0.37}\text{Ga}_{0.35}\text{Zn}_{0.28}$ as the optimal balance point, with predicted mobility of $24.3 \text{ cm}^2/\text{V}\cdot\text{s}$, threshold voltage of 1.52 V, subthreshold swing of 0.42 V/decade, and on/off ratio of 3.2×10^7 . This composition represents a Pareto-optimal solution that simultaneously satisfies the competing requirements of high-speed switching, normally-off operation, and steep subthreshold characteristics essential for next-generation AMOLED display backplanes. The framework can reduce experimental trial-and-error iterations by an estimated 80%, accelerating the materials discovery cycle from months to weeks.

Several limitations of this work warrant acknowledgment. First, the model's performance degrades at extreme composition boundaries (e.g., high-mobility regime $>32 \text{ cm}^2/\text{V}\cdot\text{s}$, quaternary Hf-doped systems) due to limited training data coverage in these regions. Future work should prioritize targeted data acquisition through active learning strategies that intelligently select the most informative compositions for experimental validation. Second, our feature set does not capture microstructural factors such as grain boundaries, interfacial roughness, or defect chemistry details that can significantly impact device performance, particularly for solution-processed or low-temperature fabricated films. Incorporating advanced descriptors from atomic-scale simulations or integrating computer vision analysis of microscopy images could address this gap.

Third, the current model assumes quasi-static device characteristics and does not predict reliability metrics such as bias-stress stability, which is critical for commercial display applications. Extending the framework to time-dependent degradation mechanisms through physics-informed neural networks that embed rate equations for defect generation and charge trapping represents an important avenue for future research. Fourth, the model is trained predominantly on sputtered films with SiO_2 gate dielectrics; generalization to alternative deposition methods (ALD, solution processing) or high- κ dielectrics requires domain adaptation techniques or transfer learning from related datasets.

Despite these limitations, this work establishes a replicable methodology for integrating heterogeneous materials data sources to accelerate functional materials discovery. The validated physics-based interpretability of learned relationships bridges the gap between black-box machine learning and scientific understanding, fostering trust and adoption in the research community. The framework is immediately applicable to other oxide semiconductor systems (e.g., SnO_2 , In_2O_3 -based TFTs) and can be extended to organic semiconductors or 2D materials with appropriate feature engineering.

From a broader perspective, this study demonstrates that materials informatics approaches can achieve predictive accuracy comparable to or exceeding experimental precision, even with modest dataset sizes (358 training instances), by strategically leveraging multi-fidelity data sources and physics-aware architectures. The combination of high-throughput DFT calculations for compositional space coverage, TCAD simulations for device-level context, and experimental data for ground-truth validation exemplifies a practical pathway toward autonomous materials discovery laboratories where AI models guide synthesis and characterization in closed-loop optimization.

Future directions include: (1) experimental validation of the top-10 predicted compositions to verify model accuracy and refine predictions through Bayesian updating; (2) extension to multi-objective evolutionary algorithms for exploring non-intuitive quaternary and quinary oxide compositions beyond human chemical intuition; (3) development of uncertainty quantification frameworks using ensemble methods or Bayesian neural networks to provide confidence intervals for predictions, enabling risk-aware decision making; (4) integration with robotic materials synthesis platforms for autonomous experimentation; and (5) open-source release of the curated dataset and trained models to accelerate community-wide progress in oxide semiconductor research. By combining domain expertise with data-driven discovery, this framework paves the way for rational design of next-generation electronic materials.

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Use unnumbered first level headings for the acknowledgments. All acknowledgments go at the end of the paper before the list of references. Moreover, you are required to declare funding (financial activities supporting the submitted work) and competing interests (related financial activities outside the submitted work). More information about this disclosure can be found at: <https://neurips.cc/Conferences/2023/PaperInformation/FundingDisclosure>.

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A Appendix / supplemental material

Optionally include supplemental material (complete proofs, additional experiments and plots) in appendix. All such materials **SHOULD be included in the main submission**.

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